

Anticipating demand in carbon-aware scheduling across data centres: guarantees and fair carbon attribution

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Abstract

Shifting deferrable jobs to cleaner times and places cuts data-centre emissions, but online schedulers act before future grid conditions and jobs are known. We formulate multi-site, deadline-constrained scheduling of bag-of-tasks jobs as a minimum-cost flow problem and study CARMA (Carbon-aware Anticipatory Receding-horizon Multi-site Allocation), which adds “ghost jobs” for expected arrivals to each hourly plan. Uncertainty about demand alone forces any online algorithm, even with perfect carbon forecasts, to a competitive ratio of $\Omega(\sqrt{\theta})$, where θ is the ratio of highest to lowest unit emission cost. Under explicit assumptions (a backstop resource, episodes within the planning window), CARMA never misses a deadline and is optimal under exact predictions; these assumptions do not hold in the weekly experiments. On 2024 regional grid data for Great Britain, CARMA comes within 8.0–9.8% of the clairvoyant optimum, against 15.5–16.3% for the best baseline without demand information. On four further fleets covering Europe, North and South America and Oceania, its gap is 2.7–5.8% against 4.4–13.0%. A scenario-based stochastic lookahead with the same demand information is level with CARMA or at most 1 point better, but needs 12–37 times more computing time. A trace-derived workload with a biased demand profile gives 21.2% against 27.5%. Under a stylised marginal-emission proxy, however, achievable reductions shrink to 14% and CARMA’s edge becomes small. Dual prices of the pooled problem give a carbon attribution in the core of the hindsight pooling game, whereas origin- and host-based accounting violate coalitional stability in every week.

Keywords: Carbon-aware computing, Data centre scheduling, Competitive

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1. Introduction

Data centres are among the fastest-growing electricity consumers. Efficiency gains kept their energy use nearly flat through the 2010s even as computing output grew several-fold [1, 2]. That buffer is now thinning, and information and communication technology accounts for an estimated 2.1–3.9% of global greenhouse-gas emissions [3]. Large machine-learning workloads have drawn particular attention [4, 5, 6]. The carbon intensity of grid electricity varies by a factor of ten or more across regions and hours, driven by the output of wind, solar and dispatchable plant [7]. Operators can therefore cut emissions without new hardware by moving flexible work to cleaner places and times. Such *carbon-aware computing* [8, 9] is an operations problem: capacity is finite, deadlines bind, and both the grid and the job stream are uncertain.

Batch jobs such as model training, analytics, rendering and backups are a natural target, because they tolerate some delay. Previous work has shown the potential of temporal shifting (running later) [10, 11, 12] and of spatial shifting (running elsewhere) [13, 14]. It has also shown their limits. Sukprasert et al. [15] found that practical reductions from spatio-temporal shifting fall well short of idealised bounds. Most methods are evaluated with a scheduler that reacts to the jobs it has already seen, using point forecasts of carbon intensity. Three questions have received less attention. First, how much is lost because an online scheduler does not know which jobs will arrive next, and can any scheduler guarantee good performance in the worst case? Second, does it pay to price the uncertainty of carbon-intensity forecasts into scheduling decisions? Third, when several sites pool their capacity, how should the resulting emissions be attributed so that every participant is better off than on its own?

This paper addresses these questions with a scheduling policy, guarantees under explicitly stated assumptions, and a controlled evaluation on three years of regional grid data for Great Britain. Its building blocks (receding-horizon planning, network flows, conformal calibration and dual cost allocation) are standard; the contribution lies in their combination for this problem, the lower bound, the conditions under which the guarantees hold and the empirical evidence. It makes five contributions.

- 35 1. **Formulation.** We state multi-site, deadline-constrained, width-limited
36 scheduling of deferrable bag-of-tasks jobs, including the emissions of
37 data transfer, as a minimum-cost flow problem (Proposition 1). Offline
38 clairvoyant optima and every online planning step can therefore be
39 solved to optimality, after quantisation, by fast combinatorial min-cost
40 flow algorithms. We also prove that myopic receding-horizon scheduling
41 can be arbitrarily worse than the clairvoyant optimum even with perfect
42 carbon forecasts (Proposition 2).
- 43 2. **Algorithm.** We propose CARMA (Carbon-aware Anticipatory
44 Receding-horizon Multi-site Allocation). At every hour, CARMA
45 adds *ghost jobs* to its planning network. These represent expected
46 arrivals, learned from the operator’s history, so the planner reserves
47 scarce low-carbon capacity for work that has not yet arrived. The
48 planning problem stays a single min-cost flow and decisions take tens
49 of milliseconds.
- 50 3. **Guarantees.** We prove that uncertainty about demand alone forces
51 every online algorithm, even one with perfect carbon forecasts, to a
52 competitive ratio of $\Omega(\sqrt{\theta})$, about $\sqrt{\theta}/2$ for large θ , where θ is the ratio
53 of the highest to the lowest unit emission (Theorem 1). In an idealised
54 setting with a backstop resource, CARMA never misses a deadline
55 whatever its predictions, so it inherits the θ -competitiveness that every
56 on-time policy has. If, in addition, each episode fits within the planning
57 window, CARMA is optimal when its predictions of demand and carbon
58 are exact, and its excess emissions grow at most linearly in the prediction
59 errors (Proposition 3 and Theorem 2). These assumptions do not hold in
60 our weekly experiments, so we verify the theorem separately in episodes
61 that satisfy them. We also show that the dual prices of the pooled
62 scheduling network give a carbon attribution among sites that lies in
63 the core of the pooling game, so no group of sites gains by leaving
64 (Theorem 3).
- 65 4. **Forecasting and recalibration pipeline.** As a tool for the scheduling
66 study, we build a reproducible 1–24-hour carbon-intensity forecaster
67 from a public feed: direct gradient-boosted models with horizon-wise
68 normalised conformal quantiles and adaptive recalibration, which keeps
69 empirical coverage close to target under the decarbonisation drift of
70 2024. Its quantiles let us test risk-adjusted carbon pricing inside the
71 scheduler. We do not claim a new forecasting method.
- 72 5. **Evidence.** Across 52 out-of-sample weeks with three workload traces

each, three load levels, two operating regimes, a trace-derived workload from the Alibaba 2018 cluster trace and four further fleets built from open generation data for Europe, the United States, Brazil and New Zealand, we compare CARMA with seven implementable online baselines, including a revenue-management protection-level policy and a scenario-based stochastic lookahead that uses the same demand information, with clairvoyant and ablated variants, and with the clairvoyant oracle. The results separate the value of anticipating demand from the value of better carbon forecasts, show when each matters, characterise how performance degrades with prediction errors, and show that origin- and host-based carbon accounting violate coalitional stability in every test week, whereas the dual attribution satisfies it exactly for the hindsight benchmark and in over 99% of the checks for CARMA’s realised emissions, where the adjusted attribution is a heuristic outside the theorem.

The results carry an operational message. Once spatial shifting is allowed, the dominant loss comes from demand myopia, not from forecast error. Anticipating demand cuts the gap of myopic planning to the clairvoyant optimum by 51–60% in the synthetic experiments, and by 42% on a trace-derived workload whose demand profile is learned from different days. Perfect carbon foresight or risk-averse carbon pricing adds little by comparison. A scenario-based stochastic lookahead with the same demand information is level with CARMA or slightly better, but 12–37 times slower, and it does not retain the network-flow structure that our guarantees use. The advantage over the best baseline without demand information persists on four other grids, from 1.7 points in Brazil to 7.3 points in Europe. Without migration, temporal shifting alone achieves much smaller reductions, and near saturation it trades emissions against lateness. We also show the limits of the evidence. The headline reductions are attributional. Under a stylised marginal-emission proxy within the single GB market, most of them would be reallocations, and physical savings depend on low-carbon supply that would otherwise be curtailed. The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 presents the model, the forecaster, CARMA and the experimental design. Section 4 establishes the theoretical guarantees, with proofs in Appendix A. Section 5 reports results, Section 6 discusses implications and limitations, and Section 7 concludes.

109 2. Related work

110 *Carbon-aware workload shifting.* Early green data-centre schedulers matched
 111 deferrable jobs to on-site solar production, for example GreenSlot [16] and
 112 GreenHadoop [11]. Other work exploited electricity-price and carbon differ-
 113 ences across sites through geographical load balancing [17, 13] and dynamic
 114 right-sizing [18]. With public carbon-intensity feeds, attention moved to grid
 115 emissions. Wiesner et al. [10] quantified the reductions from temporal shift-
 116 ing in several regions, including Great Britain, and proposed forecast-based
 117 lowest-window scheduling. Google’s carbon-intelligent compute system sets
 118 next-day virtual capacity curves from carbon forecasts and forecasts of the
 119 fleet’s own demand [8]. It is the closest industrial precedent for using demand
 120 forecasts, but it shapes hourly virtual capacity curves per cluster rather
 121 than planning individual jobs across sites. Cucumber admits delay-tolerant
 122 work only when forecast renewable supply allows it [19], and Caribou shifts
 123 serverless functions between regions [20]. CarbonScaler varies the allocation
 124 of elastic jobs with carbon intensity [12]. Ecovisor exposes the energy system
 125 to applications [21]. Carbon Explorer weighs scheduling against investment
 126 in renewables and storage [22]. Load migration between data centres can
 127 absorb curtailed renewable generation [14], and data-centre demand response
 128 has been surveyed in [23]. Sukprasert et al. [15] bound the potential of spatio-
 129 temporal shifting across 123 regions and find that simple policies capture
 130 most of the achievable benefit. Our study complements theirs by identifying
 131 *which* information limits online schedulers when capacity at clean sites is
 132 scarce.

133 *Online algorithms and guarantees.* Worst-case analysis of carbon-aware
 134 scheduling builds on online search and one-way trading [24], for which
 135 threshold policies achieve optimal competitive ratios. Lechowicz et al.
 136 derived optimal double-threshold algorithms for the online pause-and-resume
 137 problem [25]. With CarbonClipper, they also derived learning-augmented
 138 algorithms with an optimal consistency–robustness trade-off for spatiotem-
 139 poral allocation of a single deadline-constrained workload over a network
 140 of data centres [26]. These works model uncertainty about future carbon
 141 *prices* for one job. We study many jobs that compete for finite low-carbon
 142 capacity, and we show that uncertainty about *demand* creates a distinct
 143 lower bound that carbon forecasts cannot remove. Our guarantees follow
 144 the consistency–robustness paradigm of algorithms with predictions [27].

145 CARMA’s re-solving of a deterministic plan with expected future demand
146 is an instance of the certainty-equivalent re-solving heuristics of network
147 revenue management [28]. Our contribution is not the device itself but
148 its formulation as a single flow problem for carbon-aware scheduling, the
149 guarantees and the empirical attribution of value. Table 1 positions the
150 closest work along the dimensions that matter here.

Table 1: Positioning against the closest work, as modelled in the cited studies ($\checkmark$: explicitly modelled or provided; $-$: not modelled).

Work	Many jobs share finite capacity	Information on future demand	Worst-case guarantee	Attribution among sites
Lowest-window temporal shifting [10]	–	–	–	–
Geographical load balanc- ing [13]	$\checkmark$	–	–	–
Carbon-intelligent compute system [8]	$\checkmark$	$\checkmark$	–	–
Online pause-and- resume [25]	–	–	$\checkmark$	–
CarbonClipper [26]	–	–	$\checkmark$	–
Flow models of preemptive scheduling [29, 30]	$\checkmark$ (offline)	–	–	–
This paper	$\checkmark$	$\checkmark$	$\checkmark$ (A1–A2)	$\checkmark$

151 Our fairness result draws on cooperative game theory: linear production
152 games and flow games have non-empty cores that can be computed from dual
153 prices [31, 32].

154 *Energy-aware scheduling in operations.* Energy-efficient scheduling is a mature
155 topic in manufacturing operations [33], and green scheduling models have ap-
156 peared in this journal, for example for parallel machines with job splitting [34]
157 and for cloud resource allocation [35]. Most such models are deterministic and
158 solved offline, often with metaheuristics. We instead exploit exact network-
159 flow structure and embed it in an online lookahead policy. Network-flow
160 formulations of preemptive scheduling with release dates and deadlines go
161 back to Horn [29] and Federgruen and Groenevelt [30]. Proposition 1 extends
162 this classical construction to multiple sites, job widths, time-varying emission
163 costs and transfer emissions [36, 37]. In the taxonomy of Powell [38], CARMA
164 is a deterministic lookahead policy with a modified (anticipatory) model, in
165 the spirit of model predictive control [39, 40].

166 *Carbon-intensity forecasting.* Forecasts of grid carbon intensity use statistical
167 decomposition [41], machine learning with many exogenous drivers [42], day-
168 ahead regression for building demand response [43], and hierarchical models of
169 the generation mix, as in CarbonCast [44]. For Great Britain, NESO publishes
170 regional forecasts through its Carbon Intensity API [45]. These are average
171 (attributional) intensities. Marginal emission factors, which describe the plant
172 that responds to a change in load, can differ substantially from averages and are
173 the relevant measure for the system-level effect of load shifting [46, 47]. These
174 studies focus on point accuracy. For decisions, calibrated uncertainty matters
175 as well. Conformal prediction [48, 49, 50] provides distribution-free intervals
176 and has been extended to quantile regression [51], to multi-horizon [52] and
177 dynamic time series [53], and to arbitrary distribution shift through adaptive
178 conformal inference [54]. We use these tools to build a recalibrated forecaster
179 and to test, in a controlled way, whether risk-adjusted carbon costs improve
180 scheduling. The question is motivated by the optimizer’s curse [55] and by the
181 robust-optimisation view that point-estimate solutions can be fragile [56, 57].

182 3. Material and methods

183 3.1. Problem setting

184 We consider an operator that runs a set $\mathcal{S}$ of data-centre sites connected
185 by a wide-area network. Time is divided into hourly slots $t = 0, 1, 2, \dots$.
186 Deferrable batch jobs arrive online. Job j is released at the start of hour a_j
187 at origin site $o_j \in \mathcal{S}$. It carries w_j server-hours of work made of independent
188 tasks (a bag of tasks), can use at most m_j servers concurrently, and must
189 finish before the start of hour $d_j > a_j$. Work may be paused and resumed,
190 and any part of it may run at a site other than its origin. Remote execution
191 requires moving input data. At site s and hour t , $K_s(t)$ servers are free for
192 batch work once the interactive load has been served.

193 Let $I_s(t)$ be the average carbon intensity of the electricity supplied to site
194 s during hour t (gCO₂/kWh). Executing one server-hour of job j at site s
195 during hour t emits

$$c_{o_j s}(t) = e \text{PUE}_s I_s(t) + \mathbf{1}[s \neq o_j] \delta \eta \frac{1}{2} (I_{o_j}(t) + I_s(t)), \quad (1)$$

196 where e is the IT energy per server-hour, PUE_s is the power usage effectiveness
197 of site s , δ is the data volume moved per migrated server-hour and η is the
198 energy intensity of network transfer (kWh/GB). The transfer term charges

199 network energy at the mean intensity of the two endpoints. With decision
 200 variables $x_{js}(t) \geq 0$ (server-hours of job j run at site s in hour t), the offline
 201 version of the problem is the linear programme

$$\min_{x, u \geq 0} \sum_j \sum_s \sum_{t=a_j}^{d_j-1} c_{o_js}(t) x_{js}(t) + \sum_j \Pi u_j \quad (2)$$

$$\text{s.t.} \quad \sum_s \sum_{t=a_j}^{d_j-1} x_{js}(t) + u_j = w_j \quad \forall j, \quad (3)$$

$$\sum_s x_{js}(t) \leq m_j \quad \forall j, t \in [a_j, d_j), \quad (4)$$

$$\sum_{j: a_j \leq t < d_j} x_{js}(t) \leq K_s(t) \quad \forall s, t, \quad (5)$$

202 where u_j is work left unfinished at the deadline and Π is a lateness penalty
 203 much larger than any emission cost. When migration is not allowed, $x_{js}(t) = 0$
 204 for $s \neq o_j$; these variables are removed from the problem. With complete
 205 knowledge of arrivals and intensities, the optimum of (2)–(5) is a lower bound
 206 on the emissions of every schedule that serves all jobs by their deadlines, and
 207 we use it as the *clairvoyant oracle*. When emissions are evaluated only for a
 208 subset of jobs (Section 3.5), the oracle still has to serve every job under the
 209 same constraints but is charged emissions only for the evaluated ones, so that
 210 it remains a lower bound on the evaluated emissions of any schedule that
 211 meets all deadlines. Online policies that finish some work late lie outside this
 212 set; Section 3.5 explains how we compare them. An online scheduler has to
 213 fix $x_{js}(t)$ at the start of hour t . At that point it knows the jobs released so
 214 far, the current intensities $I_s(t)$ and nothing more than forecasts of $I_s(t+h)$
 215 and of future arrivals.

216 **Proposition 1** (Network-flow structure). *Problem (2)–(5) is a minimum-*
 217 *cost flow problem on a network with a source, one node per job, one node*
 218 *per (job, hour) pair, one node per (hour, site) pair and a sink. The source*
 219 *supplies $\sum_j w_j$ units of flow and the sink absorbs them. Arc capacities are w_j*
 220 *(source $\rightarrow j$), m_j ($j \rightarrow (j, t)$) and $K_s(t)$ ($(t, s) \rightarrow \text{sink}$). The only arcs with a*
 221 *cost are $(j, t) \rightarrow (t, s)$, priced at $c_{o_js}(t)$, and a slack arc $j \rightarrow \text{sink}$ priced at Π .*

222 *Proof.* Because the supply equals $\sum_j w_j$ and arc source $\rightarrow j$ has capacity w_j ,
 223 every such arc is saturated, and conservation at the job node is Eq. (3). The
 224 capacity of arc $j \rightarrow (j, t)$ is Eq. (4), conservation at (j, t) splits that hour's
 225 work across sites, and the capacity of $(t, s) \rightarrow \text{sink}$ is Eq. (5). Conversely,
 226 every feasible flow defines a feasible (x, u) with the same cost. The constraint

227 matrix of the flow formulation is a node-arc incidence matrix, in which every
 228 column has exactly one $+1$ and one -1 , so it is totally unimodular. $\square$

229 Proposition 1 has a practical consequence. The oracle and every planning
 230 problem solved online (Section 3.3) can be solved by combinatorial min-cost
 231 flow algorithms [36], which return integral optimal flows for integral data. We
 232 quantise work, widths and capacities to 0.01 server-hours (widths rounded up,
 233 capacities down) and costs to 0.01 gCO₂, and solve the quantised network
 234 to optimality with the min-cost flow solver of OR-Tools [58] (version 9.15,
 235 a cost-scaling push-relabel algorithm). The solutions are therefore optimal
 236 for the quantised problem, and the simulator treats remaining work below
 237 0.05 server-hours as complete; it checks every executed allocation against the
 238 capacity and width limits up to this tolerance. On the weekly clairvoyant
 239 problems (1,100–2,800 jobs), the flow solver returned the same objective as
 240 the LP solver HiGHS on the unquantised problem to within a relative 5×10^{-5}
 241 and was 1.4–3.5 times faster (median 2.3; 0.3–1.2 s against 0.4–3.8 s; script
 242 `bench_solvers.py`; one CPU core). The flow time includes two solves, for
 243 the emissions oracle and for the service-adjusted oracle, whereas the LP time
 244 is for one solve.

245 3.2. Probabilistic carbon-intensity forecasting

246 *Point model.* For each horizon $h = 1, \dots, H$ ($H = 24$ h) we fit a direct
 247 gradient-boosted regression tree model [59] that predicts the change $I_s(t +$
 248 $h) - I_s(t)$ from information available at issue time t . Direct per-horizon
 249 models avoid the error accumulation of recursive strategies [60]. One model
 250 per horizon is pooled across sites, with the site as a categorical feature. The
 251 features are the current intensity; differences between the current value and
 252 lags of 1, 2, 3, 6, 12 and 23 hours; the value observed at the same hour of
 253 the target on the previous day and in the previous week; a 24-hour rolling
 254 mean and mean absolute change (volatility); the national (GB) intensity; the
 255 regional wind, solar and gas shares; and calendar encodings of the target
 256 time (hour of day, day of week, annual sine and cosine). Models are fitted
 257 with absolute-error loss, so they estimate the conditional median. We use
 258 scikit-learn’s histogram gradient boosting [61] with 400 iterations, learning
 259 rate 0.05, 31 leaves and at least 50 samples per leaf. The models use only
 260 information available when the forecast is issued (gaps in the data are filled
 261 causally; Section 3.5), and no weather forecasts. This is deliberate: the goal
 262 is a forecaster any operator can build from a public intensity feed.

263 *Conformal upper quantiles.* A point forecast $\hat{I}_s(t+h|t)$ is turned into cali-
 264 brated upper quantiles by split conformal prediction [48, 49] with normalised
 265 scores. On a calibration window disjoint from training, we compute for each
 266 site s and horizon h

$$r_{s,h}(t) = \frac{I_s(t+h) - \hat{I}_s(t+h|t)}{\sigma_s(t) + \epsilon}, \quad (6)$$

267 where $\sigma_s(t)$ is the 24-hour mean absolute hourly change observed at issue
 268 time and $\epsilon = 5$ gCO₂/kWh. Only pairs whose target $t+h$ also lies in the
 269 calibration window are used, so that no calibration outcome falls in the
 270 tuning period. For a target level β , the upper forecast is $U_s^\beta(t+h|t) =$
 271 $\hat{I}_s(t+h|t) + \hat{q}_{s,h}(\beta_{s,h}(t))(\sigma_s(t) + \epsilon)$, where $\hat{q}_{s,h}(\beta)$ is the $\lceil (n+1)\beta \rceil$ -th smallest
 272 of the n calibration scores, the split-conformal quantile. Grid carbon intensity
 273 is not exchangeable over time: 2024 was markedly cleaner than 2022–2023
 274 (Fig. 3). We therefore update the working level online by adaptive conformal
 275 inference (ACI) [54]. Once $I_s(t+h)$ is observed,

$$\alpha_{s,h} \leftarrow \alpha_{s,h} + \gamma((1-\beta) - \mathbf{1}[I_s(t+h) > U_s^\beta(t+h|t)]), \quad \beta_{s,h}(t) = 1 - \alpha_{s,h}, \quad (7)$$

276 with step size $\gamma = 0.005$. At every hour, the outcomes observed in that
 277 hour update the levels before the new forecasts are issued, so feedback for
 278 horizon h arrives h hours after the forecast. When the working level reaches
 279 1, the bound is set to the largest intensity of the site in the training period,
 280 a finite worst case in place of the infinite interval of ACI; when it reaches
 281 0 the bound is zero, and bounds are floored at zero. Gap-filled outcomes
 282 give no feedback. For immediate feedback and unbounded intervals, ACI
 283 guarantees that the long-run miscoverage frequency converges to $1 - \beta$ for
 284 any data sequence [54]. Our variant has delayed feedback, a finite ceiling and
 285 a quantile grid of step 0.001, so we do not claim this guarantee and use ACI
 286 as an adaptive recalibration whose coverage is checked empirically (Appendix
 287 B). The level β then acts as a risk dial that remains roughly calibrated under
 288 drift, and with $\beta = 0.5$ the procedure removes level shifts in the point model
 289 (a recalibrated median).

290 3.3. CARMA: carbon-aware anticipatory receding-horizon multi-site allocation

291 A standard receding-horizon (model predictive control, MPC) sched-
 292 uler [39] solves, at every hour t , the flow problem of Proposition 1 restricted
 293 to pending jobs and the window $[t, t+H)$. Future intensities are replaced by

forecasts, and only the first hour of the plan is executed. Such a planner is *myopic about demand*: it sees only jobs that have already arrived. The next result shows that this can be arbitrarily costly even when carbon intensities are known exactly.

Proposition 2 (Myopia can be unboundedly costly). *Even with perfect intensity forecasts, myopic receding-horizon scheduling has no finite competitive ratio with respect to the clairvoyant oracle.*

Proof. Take a clean site A with one server and a dirty site B with ample capacity. Let $c_A(0) = \varepsilon$, $c_A(1) = 0$ and $c_B(0) = c_B(1) = c$, and ignore transfer costs. A flexible job ($w = 1$, $m = 1$) arrives at $t = 0$ with deadline $d = 2$. An urgent job ($w = 1$) arrives at $t = 1$ with $d = 2$. At $t = 0$ the myopic planner sees only the flexible job and defers it to A at $t = 1$ (cost $0 < \varepsilon$). At $t = 1$ both jobs must run in the same hour, one of them at B , so the total cost is c . The oracle runs the flexible job at $t = 0$ and the urgent job at $t = 1$, both at A , for a total of ε . The ratio c/ε is unbounded as $\varepsilon \rightarrow 0$. $\square$

The mechanism in Proposition 2 is common in practice. Low-carbon capacity is scarce, and a myopic planner spends it on flexible work that could have waited or run elsewhere, leaving urgent jobs that arrive later to run on carbon-intensive capacity. CARMA addresses it by anticipating demand inside the same flow problem, and optionally prices forecast risk.

Anticipated demand (ghost jobs). From the operator’s own history we estimate, for every hour of the week τ , origin o and window class b , the expected work $\bar{W}_{\tau ob}$ and the expected total width $\bar{M}_{\tau ob}$ of arriving jobs. Window classes are the bins $\mathcal{B} = \{1, 2, 4, 6, 9, 13, 18, 24\}$ hours. A job is assigned to the largest bin not exceeding its window, so ghost jobs are never more flexible than the jobs they represent. Because binning shortens windows, the width of each ghost job is raised where necessary to at least its work divided by its window, which makes every ghost job window-feasible. At hour t , for each $k = 1, \dots, \ell$ (the look-ahead), CARMA adds to the planning problem a *ghost job* for every (o, b) with work $\kappa \bar{W}_{\tau(t+k)ob}$, width $\kappa \bar{M}_{\tau(t+k)ob}$, release $t + k$ and deadline $\min(t + k + b, t + H)$. Ghost jobs compete with real jobs for capacity, but their slack arc is priced at a moderate penalty Π_g , set to 1.5 times the largest local unit cost in the window, rather than at Π . They can therefore reserve clean capacity when it is economical but never force real work to be late. The weight $\kappa \geq 0$ scales how much of the expected demand is anticipated. With $\kappa = 0$, CARMA reduces to myopic MPC.

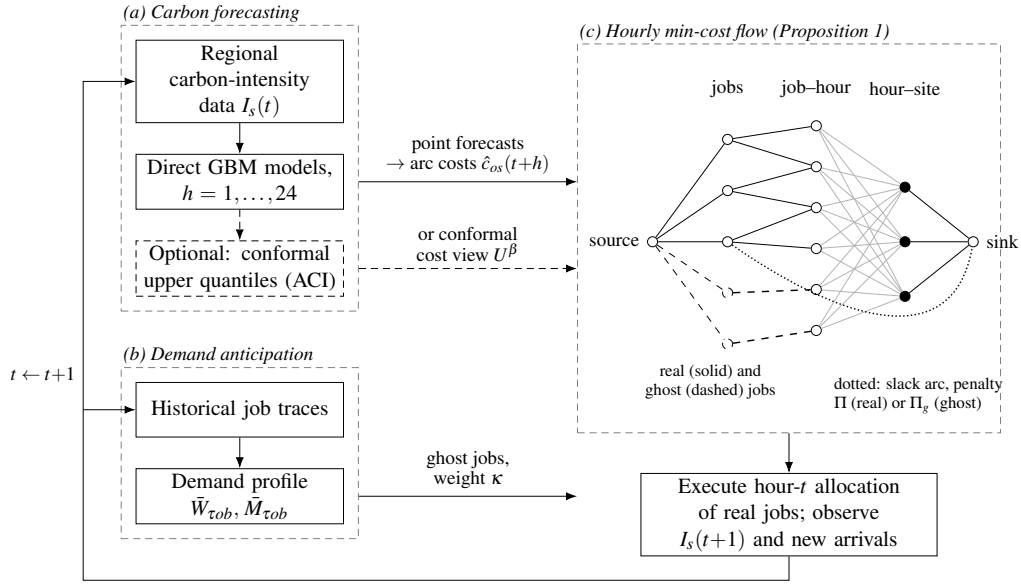


Figure 1: The CARMA decision loop. (a) Carbon-intensity forecasts price the job-hour to hour-site arcs; the selected configuration uses point forecasts, and conformal upper quantiles are an optional cost view. (b) A demand profile learned from past traces generates ghost jobs for expected arrivals. (c) Real jobs (solid) and ghost jobs (dashed) share one min-cost flow network, whose dotted slack arc carries unfinished work to the sink at the lateness penalty. The network is solved exactly every hour, and only the first-hour allocation of real jobs is executed.

330 *Carbon cost view.* Future-hour costs in Eq. (1) are evaluated at a forecast of
331 $I_s(t+h)$; for the current hour ($h=0$) the observed intensity is used. CARMA
332 admits two cost views: the point forecast $\hat{I}$, or the conformal upper quantile
333 U^β of Section 3.2. The second is motivated by the optimizer’s curse [55]:
334 selecting the minimum over noisy estimates is biased towards optimism. The
335 conformal view replaces each future cost by a calibrated quantile, which
336 shifts it by a site- and horizon-specific amount proportional to the local
337 volatility. For $\beta > 0.5$ this penalises volatile sites and distant hours more than
338 stable sites and near hours. Whether this risk premium helps is an empirical
339 question, and we treat the cost view as a hyperparameter.

340 *Algorithm.* Fig. 1 summarises the loop. At each hour t , CARMA (i) observes
341 $I_s(t)$, the free capacity and the pending jobs; (ii) builds the cost cube $c_{os}(t+h)$
342 for $h < H$ from the chosen forecast view; (iii) adds the ghost jobs for hours
343 $t+1, \dots, t+\ell$; (iv) solves the resulting min-cost flow problem exactly; (v)
344 executes the first-hour allocations of the real jobs only; and (vi) after observing
345 the realised intensities, updates the ACI levels by Eq. (7) when conformal
346 costs are used. Work that is still pending at its deadline receives a one-hour
347 planning window, which makes it urgent in the next plan; it carries the same
348 penalty as other real work, so no strict priority is imposed. The planning
349 network has $O((n_t + \ell|\mathcal{S}||\mathcal{B}|)H|\mathcal{S}|)$ arcs, where n_t is the number of pending
350 jobs, so each decision takes a few tens of milliseconds (Section 5). The
351 hyperparameters (κ, ℓ) and the cost view are tuned on validation weeks that
352 precede the test period (Section 5.2).

353 3.4. Baselines

354 We compare CARMA with the following online policies and the oracle.
355 **ASAP-Local** is carbon-agnostic: it runs work at its origin as soon as possible,
356 earliest deadline first (EDF). **Greedy-Spatial** also runs work immediately,
357 but at the site that is currently cleanest; this is geographical load balancing
358 without temporal shifting [13]. **Forecast-Reserve** books, when a job arrives,
359 the cheapest forecast (site, hour) slots in its window that still have capacity,
360 and never revises the booking. This is the lowest-window strategy of [10]
361 extended to multiple sites. **MPC** is the myopic receding-horizon flow scheduler
362 with point forecasts. **MPC-Protect** adds a protection level in the spirit
363 of revenue management: planned work may use only a share $1-r$ of the
364 capacity of the sites that are cleaner than the fleet median over the window.
365 In the current hour, the protected share is first offered to jobs due within τ

366 hours and then, because unused capacity perishes, to all remaining work. The
 367 parameters (r, τ) are tuned on validation data. **MPC+conformal** uses the
 368 conformal upper quantile at the level that was best for MPC on validation
 369 data. **MPC-Perfect** is MPC given the realised intensities, which isolates the
 370 value of forecast accuracy. **CARMA-Perfect** is CARMA given the realised
 371 intensities, and **CARMA+conformal** is CARMA with the best conformal
 372 cost view. **Scenario-MPC** is a demand-aware comparator that uses the
 373 same information as CARMA in a different way. At every hour it draws
 374 K arrival scenarios for the next ℓ hours from the historical job traces from
 375 which CARMA’s profile is estimated (the jobs of one historical week, or trace
 376 day, at the same hours), and solves a two-stage stochastic programme by
 377 sample-average approximation: the first-hour allocation of the pending jobs is
 378 shared, while the later allocation and the scenario jobs are planned separately
 379 in each scenario, and the mean cost is minimised. Unserved scenario work
 380 carries CARMA’s ghost penalty Π_g . The coupling across scenarios breaks the
 381 network structure, so Scenario-MPC is solved as an LP with HiGHS; (K, ℓ)
 382 are tuned on validation data. **Oracle** solves (2)–(5) with full knowledge of
 383 arrivals and intensities.

384 3.5. Data and experimental design

385 *Carbon intensity.* We use half-hourly regional carbon intensity for Great
 386 Britain from the Carbon Intensity API of the National Energy System Op-
 387 erator (NESO) [45] from 1 January 2022 to 31 December 2024. Values are
 388 averaged to hourly resolution. NESO does not meter regional intensity. For
 389 past periods, the regional endpoint returns a single modelled estimate per
 390 half-hour (in a field named `forecast`), and we treat this estimate as the
 391 realised intensity. The oracle and the “Perfect” variants are therefore clair-
 392 voyant with respect to NESO’s published regional values, and all emission
 393 figures inherit their modelling error. NESO’s forward regional forecasts are
 394 not archived, so they cannot serve as a historical benchmark for our forecaster.
 395 Four regions host the candidate sites: South Scotland, North West England,
 396 West Midlands and London. They span the north–south gradient of the
 397 British grid, from wind- and nuclear-rich South Scotland to gas-dependent
 398 London, and include the main data-centre markets of London, Manchester
 399 (North West England) and Birmingham (West Midlands). The largest cluster
 400 of the London market, around Slough, lies in NESO’s South England region;
 401 we use the London region as a stylised proxy for a southern, gas-dependent
 402 site. Our first download missed 602 of the 26,304 hours in at least one region,

most of them in two blocks of eleven to twelve days at the start of 2023 and 2024. Re-requesting these days in one-day chunks recovered all but 52 hours, which the API does not provide: 41 hours on 20–22 October 2023 (in the tuning period) and 11 hours on 11–12 June 2024. These are filled causally by carrying the last observation forward, so no filled value uses later data. Filled hours are excluded from forecast training, calibration, ACI feedback and forecast evaluation. In the scheduling simulation they act as realised intensities; the two test weeks whose episodes contain them are checked separately in Section 5.3. The period up to 30 June 2023 is used to train the forecaster, July–September 2023 to calibrate the conformal scores, October–December 2023 to tune the scheduler, and all 52 weeks starting on Mondays in 2024 as the out-of-sample test set.

Other grids and fleets. To test generality, we build four further fleets of four sites from open generation-mix data for 2022–2024 (Table 2; script `grids_fetch.py`). Hourly carbon intensity is the generation-weighted mean of the IPCC (2014) lifecycle emission factors of the fuels in the mix (coal 820, gas 490, oil 650, biomass 230, nuclear 12, hydro 24, wind 11, solar 45, geothermal 38, waste 400 and other 500 gCO₂e/kWh). This is a production-based measure: imports, exports and storage are not accounted for, and its definition differs from the direct-emission factors that NESO applies to Great Britain, so absolute levels are not comparable across fleets. Only the ratios between sites of one fleet matter for scheduling. Sites within a fleet are ordered from the cleanest to the dirtiest mean intensity; capacity (300, 400, 400 and 600 servers) and job-origin shares (10, 20, 25 and 45%) follow this order as in Great Britain, and a reversed layout is tested. PUE is 1.2 at all sites. The arrival rate of each origin and the interactive load follow the local standard time of its site. Gaps in the data (4 to 124 hours per fleet) are filled causally as for Great Britain. Open hourly generation data for Asia and Africa could not be retrieved (for example, Eskom publishes only the last seven days), so these regions are not covered.

Sites and energy parameters. For Great Britain, batch capacities are 300, 400, 400 and 600 servers for South Scotland, North West England, West Midlands and London. This reflects a small, clean northern site and a large site in London; the split is varied in the sensitivity analysis. Interactive load takes $20\% \pm 15\%$ of each site’s capacity on a diurnal cycle that peaks in the afternoon. We assume $e = 0.4$ kWh per server-hour. The assumed PUE values are 1.15,

Table 2: Fleets. Mean carbon intensity 2022–2024 in gCO₂e/kWh (lifecycle factors; Great Britain: NESO regional values) and UTC offset of local standard time in hours.

Fleet	Sites (mean intensity; UTC offset)	Source
Great Britain	South Scotland (37), North West England (58), West Midlands (156), London (163); 0	NESO Carbon Intensity API
Europe	France (52), Spain (156), Germany (335), Poland (598); +1	Energy-Charts
United States	BPA (64; −8), California (256; −8), Texas (337; −6), PJM (353; −5)	EIA-930
Brazil	Nordeste (39), Sudeste/CO (72), Sul (98), Norte (126); −3	ONS
Four continents	France (52; +1), New Zealand (98; +12), Brazil South (98; −3), California (256; −8)	as above; NZ Electricity Authority

1.20, 1.25 and 1.30 from north to south, reflecting free-cooling potential. Migrated work moves $\delta = 2$ GB per server-hour at $\eta = 0.02$ kWh/GB. This value lies within the wide range of published estimates of transmission energy intensity [62]. The main experiments count only the energy of busy servers, which assumes that idle servers are switched off or reallocated. The sensitivity analysis varies η and the capacity split, and it adds an always-on case with an idle draw of 0.2 kWh per server-hour for every batch server.

Workloads. Job arrivals follow a non-homogeneous Poisson process with a diurnal peak in the afternoon and 30% lower weekend rates. Jobs originate at the four sites with probabilities 0.10, 0.20, 0.25 and 0.45. Width m_j is log-uniform on $[4, 64]$ servers and the run time at full width r_j is uniform on $\{1, \dots, 6\}$ hours, so $w_j = m_j r_j$. A share v of jobs is urgent (deadline $= a_j + r_j$, no temporal slack); for the others the window is $\lceil f_j r_j \rceil$ hours with $f_j \sim U[1.5, 4]$, capped at 24 hours. The baseline value is $v = 0.3$. The arrival rate is scaled so that the offered load equals a fraction $\rho \in \{0.3, 0.5, 0.7\}$ of the mean free capacity of the fleet. Each test episode covers one week of arrivals. We also simulate 24 hours of arrivals before and after the week, so the evaluated week starts and ends under steady-state congestion. Metrics are computed only for jobs released during the week. For each week and load level we draw three independent workload traces with fixed seeds, and every policy runs on the same traces. The demand profile used by CARMA is estimated from 13 independent historical traces of the same generator. This favours anticipation, because the profile is then an unbiased estimate of the

462 arrival rate. We therefore also evaluate a trace-driven workload whose profile
 463 is learned from different days (next paragraph) and profiles learned at the
 464 wrong load (Section 5.5). We study two regimes: *spatio-temporal*, where
 465 migration is allowed, and *temporal-only*, where it is not. All hyperparameters
 466 are tuned separately for each regime.

467 *Trace-derived workload.* To test anticipation on real, non-stationary demand,
 468 we derive a fluid workload from the batch-task table of the Alibaba 2018
 469 cluster trace [63, 64]. The table has 14.3 million task records over eight
 470 days. We keep the 13.3 million terminated tasks with valid timestamps and a
 471 planned CPU request and group them by job (one directed acyclic graph of
 472 tasks), which gives 3.85 million jobs (script `prepare_trace.py`). The trace
 473 records no submission times, so a job’s arrival is the start of its first task. Its
 474 work is the sum over tasks of instances $\times$ run time $\times$ planned CPU request
 475 (core-hours), and its width is the largest planned core count of a single task;
 476 planned requests are not measured utilisation. Both are converted to servers
 477 of 96 cores and scaled so that the offered load in the evaluation days equals
 478 $\rho = 0.5$. Task precedence within a job is ignored. The trace records no
 479 deadlines, so windows are drawn as in the synthetic model: 30% of jobs are
 480 urgent, with a window equal to their run time rounded up to whole hours,
 481 and the others receive 1–23 hours of slack. Origins are assigned at random
 482 with the site probabilities above. To keep the planning problems tractable,
 483 jobs are aggregated into batches by arrival hour, origin and window class,
 484 and the works and widths of a batch are added. Pooling widths relaxes
 485 the per-job width limits, because a batch may run its short, wide members
 486 and its long, narrow members at a combined rate that no single member
 487 could sustain. The batched workload is therefore an approximation, and we
 488 test its effect with finer batches that also separate jobs by run-length class
 489 (Section 5.4). CARMA and Scenario-MPC learn from days 1–4 of the trace
 490 and are evaluated on days 5–8, preceded by 24 hours of warm-up. Days
 491 5–8 carry 39% more work than days 1–4, so the learned hour-of-day profile
 492 underestimates evaluation demand by 28%, and hourly work ranges from 0.2
 493 to 3.6 times its mean, so bursts are not anticipated. The four evaluation days
 494 are paired with each of the 52 test weeks of 2024, starting on the Monday,
 495 with three independent draws of origins and deadlines. These 156 episodes
 496 vary the carbon conditions and the synthetic attributes, but they share one
 497 demand chronology.

498 *Metrics and statistics.* The primary metric is operational emissions (tCO₂)
 499 per week under average regional intensity, which is an *attributional* measure
 500 (Section 6 discusses marginal emissions). We report the reduction relative to
 501 ASAP-Local, the gap to the clairvoyant oracle, the share of late work, the
 502 share of work executed away from its origin and the decision time. Reductions
 503 and gaps are means of per-episode ratios, so they cannot be recomputed from
 504 the mean emissions in the tables. Late work is work still outstanding when
 505 its deadline passes; online policies keep processing it, all of it is completed
 506 before the end of the simulation, and its emissions are counted. The gap to
 507 the oracle is an optimality gap only for schedules that meet all deadlines.
 508 Where lateness is negligible, as in the spatio-temporal regime, it measures
 509 the loss of an online policy against the optimum. Where it is not, emissions
 510 and service must be read together, and we also report a service-adjusted
 511 gap. It charges each late server-hour a penalty equal to the largest local
 512 unit cost of the episode, in addition to its emissions, and compares with
 513 an oracle that minimises the same objective: evaluated work may be left
 514 undone at that penalty, while boundary work must still be served on time.
 515 The on-time part of any schedule that serves its boundary work on time is
 516 feasible for this oracle, so its value bounds the adjusted emissions of such
 517 schedules from below. The estimand for comparisons is the mean over the
 518 52 test weeks of the paired difference in gap between a policy and CARMA,
 519 using week-level means over the three traces. Consecutive weeks are serially
 520 correlated (lag-1 autocorrelation of the paired differences ≈ 0.5), so confidence
 521 intervals and two-sided p -values come from a moving-block bootstrap over
 522 weeks [65] with blocks of four weeks and 5,000 draws. The p -value is the
 523 share of bootstrap means of the centred series that are at least as far from
 524 zero as the observed mean. We report block lengths of two and eight weeks as
 525 a sensitivity check. p -values are Holm-adjusted jointly over all comparisons
 526 of the main experiment (both regimes and all loads) and separately within
 527 the trace and sensitivity experiments.

528 4. Theoretical guarantees

529 This section establishes a worst-case limit for *every* online scheduler,
 530 prediction-dependent guarantees for CARMA, and a fairness property of the
 531 carbon attribution induced by the pooled scheduling problem. Proofs are
 532 in Appendix A. Throughout, unit costs (Eq. (1), including any backstop
 533 resource introduced below) satisfy $0 < L \leq c_{os}(t) \leq U$, and $\theta = U/L$ is

the *unit-emission-cost ratio*. It refers to unit costs, which include energy use, PUE, transfer overheads and any backstop, not to grid intensities alone. Operational costs alone can be zero, because the hourly regional intensity of South Scotland is 0 gCO₂/kWh in 133 hours of our 2022–2024 data. A positive floor appears once the embodied emissions of the servers are allocated per server-hour, which is an accounting convention rather than a claim that scheduling avoids embodied emissions. As an illustration, a floor of 30 gCO₂e added to every unit cost, with the largest operational unit cost observed in 2024 (221 g), gives $\theta = 251/30 \approx 8.4$. An online algorithm is Γ -competitive if its emissions never exceed Γ times those of the clairvoyant oracle, for any arrival sequence and any intensity trajectory.

4.1. A universal lower bound under demand uncertainty

Competitive analyses of carbon-aware scheduling model uncertainty about future *prices*, that is, intensities, for a single workload. Examples are online search and one-way trading [24], the online pause-and-resume problem [25], and spatiotemporal allocation with deadlines (CarbonClipper) [26]. When several jobs share scarce low-carbon capacity, difficulty remains even when all prices are known in advance, because future *demand* is unknown.

Theorem 1 (Lower bound). *Let $\theta > 1$. No online algorithm (deterministic or randomised, with integral or fractional decisions) can be better than $R^*(\theta)$ -competitive, even if it knows all future carbon intensities exactly, where*

$$R^*(\theta) = \max_{1 \leq p \leq \theta} \left\{ 1 + \frac{(p-1)(\theta-p)}{p^2 - p - 1 + \theta} \right\} \geq 1 + \frac{\sqrt{\theta}(\sqrt{\theta}-1)^2}{2\theta - \sqrt{\theta} - 1} = \Omega(\sqrt{\theta}). \quad (8)$$

On the same family of instances, the competitive ratio of myopic receding-horizon scheduling (MPC) is at least $(1 + \theta)/2$, a value approached as $p \rightarrow 1$.

For $\theta = 1$ every schedule costs the same and $R^*(1) = 1$; both expressions in (8) tend to 1 as $\theta \downarrow 1$.

The bound is established on a two-hour instance and is a quantitative version of Proposition 2, which treats the case $L = 0$. Any algorithm must decide how much flexible work to run now at moderate cost before it learns whether an urgent job will claim the cheap capacity later. The adversary thus raises the effective price of the clean hour from L to U . The resulting order, $\sqrt{\theta}$, is the same as for deterministic online min-search with unknown prices [24]. The difference is where it comes from: here every price is known,

566 and the loss is caused only by the unknown arrival of a competing job. For
 567 the illustrative $\theta \approx 8.4$ above, $R^* = 1.81$, whereas myopic MPC can be driven
 568 to 4.7. For $\theta = 100$ the values are 5.29 and 50.5. $R^*(\theta)$ is a lower bound
 569 derived from one family of instances; we do not know whether any online
 570 algorithm attains it. Theorem 1 is a worst-case statement, but it points
 571 in the same direction as the average-case evidence of Section 5.3. Carbon
 572 forecasts cannot substitute for information about demand, and this motivates
 573 the learning-augmented design of CARMA [27].

574 4.2. Deadline safety, consistency and smoothness of CARMA

575 We analyse CARMA under two assumptions.

576 (A1) *Backstop capacity.* In every hour, work may also run on a backstop
 577 resource of unlimited capacity at a unit cost in $[L, U]$, for example
 578 by bursting to a public-cloud region or to reserve servers at a carbon-
 579 intensive site. The backstop’s unit cost is known in advance, or its
 580 forecast is clipped to $[L, U]$, so the planner never prices it above U . All
 581 jobs, real and predicted, are *window-feasible*, $w_j \leq m_j(d_j - a_j)$, and
 582 have windows of at most H hours, so every planning problem contains
 583 all pending deadlines (this holds by construction in our workloads). The
 584 penalties satisfy $\Pi > U$ for real jobs and $\Pi_g \geq U$ for ghost jobs. The
 585 implementation enforces window-feasible ghost jobs (Section 3.3) and
 586 sets Π_g to 1.5 times the largest local unit cost, which exceeds every unit
 587 cost including transfer overheads.

588 (A2) *Episode within the planning window.* The episode $\{0, \dots, T - 1\}$ has
 589 length $T \leq H$, as for a daily batch window, so every planning problem
 590 extends to the end of the episode.

591 To measure prediction quality at hour t , let F_t be the set of jobs that actually
 592 arrive after t , and $\hat{F}_t$ the ghost jobs used by CARMA. Jobs are compared
 593 after splitting them into *atoms* with proportional widths, which leaves every
 594 planning problem unchanged (Appendix A). Let d_t be the smallest total
 595 work of atoms that must be added to or removed from $\hat{F}_t$ to obtain F_t . Let
 596 $\varepsilon_t = \max_{\tau > t, o, s} |\hat{c}_{os}(\tau | t) - c_{os}(\tau)|$ be the carbon-cost forecast error, and W_t
 597 the total work pending at t plus the larger of the true and predicted future
 598 work.

599 **Proposition 3** (Baseline robustness). *Under (A1), every policy that completes*
 600 *all work by its deadlines is θ -competitive.*

601 **Theorem 2** (Guarantees for CARMA). *Under (A1), for any arrival sequence*
602 *and any predictions, CARMA completes every job by its deadline (deadline*
603 *safety) and is therefore θ -competitive. If (A2) also holds, then*

$$\text{ALG} \leq \min \left\{ \theta \cdot \text{OPT}, \text{OPT} + \sum_{t=0}^{T-1} [(U - L) d_t + 2 \varepsilon_t W_t] \right\}. \quad (9)$$

604 *In particular, CARMA is optimal with exact predictions ($d_t = \varepsilon_t = 0$), which*
605 *is consistency, and its excess emissions grow at most linearly in the prediction*
606 *errors, which is smoothness.*

607 These properties should be read with care. Robustness by itself does not
608 distinguish CARMA: by Proposition 3 it holds for any policy that meets
609 its deadlines. What CARMA adds is deadline safety under *arbitrary* ghost
610 predictions, together with consistency and smoothness. Deadline safety holds
611 because ghost jobs carry a softer penalty than real jobs and the backstop can
612 always absorb real work, so a misprediction can divert real work to dirtier
613 capacity but never make it late. Without a backstop this is no longer true.
614 With one server and two hours, a flexible job that could run in the first hour
615 may be deferred to a cleaner second hour, and an unexpected urgent job in
616 the second hour then makes one of them late, although an offline schedule
617 meets both deadlines. A soft ghost penalty does not create capacity. “Exact
618 predictions” in the theorem means exact predictions of both demand and
619 carbon; the CARMA-Perfect variant of the experiments has perfect carbon
620 forecasts but still uses the average demand profile and a finite look-ahead,
621 so it is not the case $d_t = 0$. The smoothness term charges demand errors
622 at the spread $U - L$ of unit costs, not at the lateness penalty. The term is
623 informative when predictions are close to the realised demand. For random
624 arrivals, however, an expected-value profile never matches the realised jobs
625 exactly. Moreover, each misprediction enters d_t in every epoch in which it is
626 visible, so the bound in (9) can be loose. In the main experiments, assumption
627 (A2) is not satisfied (weekly episodes with $H = 24$ h), no backstop is modelled,
628 and some hours have zero operational cost. The theorem therefore describes
629 an idealised setting and does not certify the reported numbers; the degradation
630 profile in the weekly experiments is tested empirically in Section 5.7. Appendix
631 C checks the theorem numerically in one-day episodes that do satisfy (A1) and
632 (A2). Robustness θ is also far from the $\Omega(\sqrt{\theta})$ lower bound. Closing this gap
633 while retaining consistency is left for future work, for example by protecting

634 deferral decisions with reservation prices [24, 26]. Fig. 2 summarises the
 635 worst-case landscape.

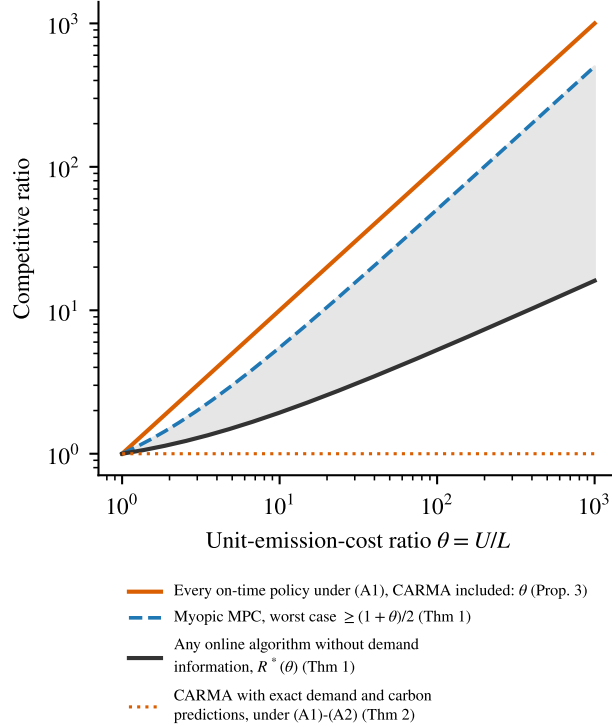


Figure 2: Worst-case landscape as a function of the unit-emission-cost ratio $\theta = U/L$. The curves refer to different information regimes. Without information about future demand, no online algorithm can lie below $R^*(\theta)$, even with exact carbon forecasts (Theorem 1), and myopic MPC can be driven to $(1 + \theta)/2$. Given a backstop (A1), every policy that meets its deadlines, CARMA included, is within θ (Proposition 3). With exact predictions of *both* demand and carbon and an episode within the planning window (A1–A2), CARMA is optimal (Theorem 2).

636 4.3. Fair carbon attribution among sites

637 Pooling capacity reduces total emissions but redistributes them. Work
 638 submitted in London is executed in South Scotland, and Scottish jobs may be
 639 displaced by it. The organisations that operate the sites, or the business units
 640 that submit the jobs, therefore need an attribution of the pooled emissions
 641 that each of them can accept. This is an *internal* allocation problem, like

642 carbon transfer pricing between units. It is distinct from external GHG
643 Protocol reporting, where Scope 2 emissions are reported by the entity that
644 consumes the electricity [66].

645 We model the allocation as a cooperative game. The players are the sites $\mathcal{S}$.
646 A coalition $C \subseteq \mathcal{S}$ pools the capacity of its sites and serves the jobs submitted
647 at them. Its stand-alone value $c(C)$ is the optimum of (2)–(5) restricted
648 to those jobs and sites, which includes the lateness penalty if the coalition
649 cannot serve its own jobs on time. An attribution $\phi \in \mathbb{R}^{|\mathcal{S}|}$ is *budget-balanced*
650 if $\sum_o \phi_o = c(\mathcal{S})$. It is in the *core* if $\sum_{o \in C} \phi_o \leq c(C)$ for every coalition C . A
651 core attribution gives no group of sites a reason to leave the pool and schedule
652 on its own.

653 Let (λ, ν, μ) be optimal dual variables of the grand-coalition problem: λ_j
654 for the work constraint (3), $\nu_{jt} \leq 0$ for the width constraint (4) and $\mu_{st} \leq 0$
655 for the capacity constraint (5). The *dual carbon attribution* is

$$\phi_o = \sum_{j: o_j=o} \left(\lambda_j w_j + \sum_t \nu_{jt} m_j \right) + \sum_t \mu_{ot} K_o(t). \quad (10)$$

656 Site o is charged the carbon shadow price of the work it submits and is
657 credited ($\mu \leq 0$) with the shadow value of the capacity it contributes. A
658 site whose clean capacity is worth more than the emissions of its own jobs
659 therefore receives a net credit, that is, a negative internal charge.

660 **Theorem 3** (Core-stable carbon attribution). *The attribution (10) is budget-*
661 *balanced and in the core of the pooling game. In particular, it is individually*
662 *rational: no site is charged more than its stand-alone optimum $c(\{o\})$.*

663 The result follows from Owen’s theorem on linear production games [31];
664 games defined by network flows are known to be totally balanced [32]. It
665 concerns the penalised hindsight game defined above. Core stability is one
666 requirement of an acceptable allocation; it does not imply uniqueness, incentive
667 compatibility or other notions of fairness, and it does not extend automatically
668 to the realised emissions of an online policy. The duals are those of the
669 hindsight problem, which suits after-the-fact accounting. They are available
670 as the optimal node potentials of the min-cost flow; in our experiments
671 we obtain them from an LP solver. Optimal duals need not be unique, so
672 Section 5.8 reports the range of (10) over all optimal duals. It also compares
673 the dual attribution with the Shapley value [67], which is budget-balanced but
674 not guaranteed to lie in the core, and with simpler rules. The simpler rules are

675 *origin-based* (each site is charged the emissions of its own jobs wherever they
676 ran), *host-based* (each site is charged the emissions of the work it executes,
677 the analogue of location-based Scope 2 reporting), and proportional rules.

678 5. Results

679 5.1. Carbon-intensity data and forecasts

680 Fig. 3 shows the regional intensities. Over 2022–2024, mean intensity
681 was 37 gCO₂/kWh in South Scotland, 58 in North West England, 156 in the
682 West Midlands and 163 in London. The regional means indicate large spatial
683 heterogeneity, and it persists hour by hour: the spread between the cleanest
684 and the dirtiest of the four sites has a median of 156 gCO₂/kWh and exceeds
685 54 g in 90% of hours. This is what makes spatial shifting attractive. Annual
686 mean intensities declined at every site; London’s fell from 202 gCO₂/kWh in
687 2022 to 125 in 2024. Diurnal profiles peak in the early evening (around 18:00
688 UTC), and the profile of South Scotland is much flatter than those of the
689 southern sites. The four series are positively correlated (pairwise correlations
690 between 0.51 and 0.87), because national wind output drives all of them.

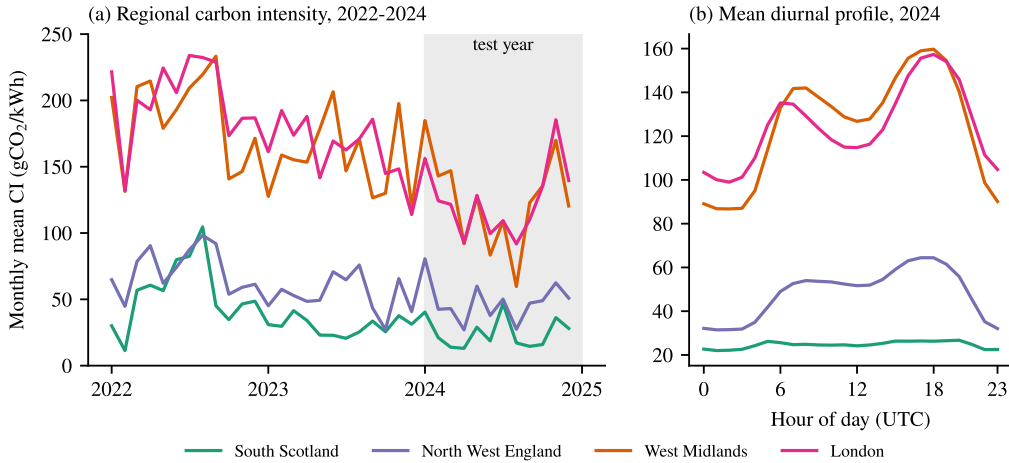


Figure 3: Hourly carbon intensity of the four candidate sites (NESO regional data). (a) Monthly means, 2022–2024; the shaded area is the out-of-sample test year. (b) Mean diurnal profile in 2024 (UTC).

691 The forecaster was trained on 1.24 million (issue time, site, horizon)
692 samples in about one minute. In 2024 its mean absolute error (MAE), averaged

over horizons and sites, is 31.7 gCO₂/kWh, against 35.3 for persistence and 40.9 for a seasonal-naive forecast. The advantage is 23–25% at horizons of 1–6 hours, and the difference is significant at all English sites up to 18 hours in Diebold–Mariano tests [68]. Day-ahead skill is limited, because no weather inputs are used. Adaptive recalibration keeps the annual coverage of the upper bounds within 0.003 of target despite the 2024 decarbonisation drift. Appendix B gives the full evaluation (accuracy by horizon and site, coverage, pinball loss).

5.2. Hyperparameter tuning

All policies were tuned on 13 validation weeks (October–December 2023, $\rho \in \{0.3, 0.5, 0.7\}$), separately for each regime. Fig. 4 shows the spatio-temporal regime. Anticipating demand helps across the whole grid: every tested configuration beats myopic MPC, whose mean gap to the oracle is 22.3%. The gap is smallest around $\kappa = 1$, where expected demand is anticipated without inflation: 9.3% for $\ell = 12$ and 9.1% for $\ell = 18$. Over-anticipation ($\kappa = 1.25$ with $\ell = 18$) reserves too much clean capacity and is penalised (12.4%). We selected $\kappa = 1$ and $\ell = 12$ hours, resolving configurations within 0.25 percentage points of the best towards the shorter look-ahead. In the temporal-only regime, anticipation brings no benefit. The best configuration ($\kappa = 0.5$, $\ell = 6$; 4.29%) is indistinguishable from MPC (4.29%), and larger κ is slightly worse. For MPC-Protect, the best protection level was $r = 0.3$ with $\tau = 1$ h in the spatio-temporal regime (21.3% against 22.3% for MPC). In the temporal-only regime every protection level was worse than MPC (at best 5.9%). For Scenario-MPC, $K = 8$ scenarios with $\ell = 12$ gave 9.0%, against 9.3% with $K = 4$ and 12.5–12.7% with $\ell = 6$.

The cost view interacts with anticipation (Fig. 4b). For myopic MPC, high conformal levels help. The gap falls from 25.5% at $\beta = 0.5$ to 17.3% at $\beta = 0.9$ and rises again at $\beta = 0.95$ (18.5%), against 22.3% with point forecasts. For CARMA, every conformal level is worse than the point forecast: the best, $\beta = 0.5$, gives 10.1% against 9.3%. The mechanism is not a uniform risk premium. On the validation weeks, the $\beta = 0.6$ view lies on average between 1.5 g below and 3.9 g above the point forecast at South Scotland, North West England and London, but 14.5 g above it at the volatile West Midlands site. The offset also grows with the horizon, and at $\beta = 0.8$ it reaches 23–95 g at a 24-hour horizon. High levels therefore make distant hours and volatile sites look dirtier, so the planner runs work now and at stable sites, which keeps future clean capacity free. For a myopic planner this is a crude

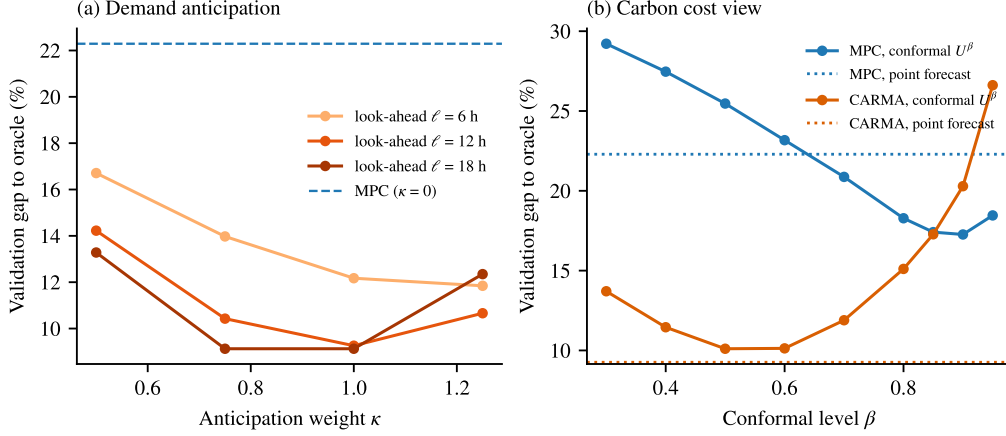


Figure 4: Validation results for the spatio-temporal regime (2023-Q4, mean over 13 weeks and three load levels). (a) Gap to the oracle as a function of the anticipation weight κ and look-ahead ℓ , using point forecasts. (b) Effect of the carbon cost view: conformal quantiles U^β against point forecasts, for myopic MPC and for CARMA ($\kappa = 1$, $\ell = 12$).

form of anticipation. Once CARMA represents future demand explicitly, the same distortion double-counts it and gives up useful deferrals. The selected CARMA therefore uses point forecasts. For the test we keep MPC+conformal at $\beta = 0.9$ and CARMA+conformal at $\beta = 0.5$ (spatio-temporal) as ablations. In the temporal-only regime, the best conformal views were $\beta = 0.6$ for MPC and $\beta = 0.5$ for CARMA.

5.3. Out-of-sample scheduling performance

Tables 3 and 4 report results for the 52 test weeks of 2024, with three independent workload traces per week. Weekly traces contain on average 917, 1,527 and 2,141 jobs (69,400, 115,400 and 162,000 server-hours) at $\rho = 0.3$, 0.5 and 0.7. Fig. 5 shows the gaps to the oracle.

Spatio-temporal regime. With migration allowed, the oracle cuts attributional emissions relative to ASAP-Local by 71.8%, 58.2% and 41.1% at $\rho = 0.3$, 0.5 and 0.7. Greedy-Spatial realises only part of this potential (gaps of 38–67%), because it has no temporal flexibility and floods the cleanest site with whatever arrives first. The forecast-driven planners without demand information come closer. Forecast-Reserve has gaps of 24–27% and MPC 19–23%. MPC+conformal (15.5–16.6%) is the best of them at $\rho = 0.3$ and

Table 3: Out-of-sample results in the spatio-temporal regime (migration allowed), 52 weeks of 2024×3 traces. Emis.: mean weekly operational emissions (tCO₂, attributional). Red.: reduction relative to ASAP-Local (%). Gap: excess over the clairvoyant oracle (%). Adj.: service-adjusted gap, in which late work is additionally charged at the episode’s largest local unit cost and the oracle minimises the same objective. Late: share of work outstanding at its deadline (%). Mig.: share of work executed away from its origin (%). ms: mean decision time per hour (one CPU core). p : two-sided moving-block bootstrap test of the mean paired weekly difference in gap against CARMA, Holm-adjusted jointly over all comparisons in Tables 3 and 4. Red. and Gap are means of per-episode ratios. The best gap among implementable online policies is in bold.

Load	Policy	Emis.	Red.	Gap	Adj.	Late	Mig.	ms	p
$\rho = 0.3$	ASAP-Local	3.71	0.0	284.0	284.0	0.00	0.0	0.03	<0.001
	Greedy-Spatial	1.72	54.7	66.8	66.8	0.00	81.8	0.10	<0.001
	Forecast-Reserve	1.34	64.9	26.4	26.4	0.00	83.8	0.39	<0.001
	MPC	1.28	66.8	19.1	19.1	0.00	69.5	2.1	<0.001
	MPC-Protect	1.37	64.2	28.8	28.8	0.00	80.5	1.8	<0.001
	MPC+conformal	1.24	67.7	15.5	15.5	0.00	70.0	1.9	<0.001
	MPC-Perfect	1.25	67.5	16.7	16.7	0.00	69.1	2.1	<0.001
	Scenario-MPC	1.18	69.4	9.0	9.0	0.00	70.2	248	0.019
	CARMA	1.18	69.3	9.3	9.3	0.00	70.1	21	–
	CARMA+conformal	1.18	69.2	9.9	9.9	0.00	70.3	21	0.185
	CARMA-Perfect	1.12	71.1	2.5	2.5	0.00	69.9	21	<0.001
	Oracle	1.09	71.8	0.0	0.0	0.00	70.9	–	–
$\rho = 0.5$	ASAP-Local	6.11	0.0	149.6	150.0	0.05	0.0	0.05	<0.001
	Greedy-Spatial	4.11	33.6	63.1	63.1	0.00	76.5	0.17	<0.001
	Forecast-Reserve	3.29	47.6	26.8	27.0	0.02	78.3	0.65	<0.001
	MPC	3.19	49.2	23.0	23.1	0.01	50.1	4.2	<0.001
	MPC-Protect	3.05	51.7	16.3	16.3	0.00	58.8	4.9	<0.001
	MPC+conformal	3.05	51.6	16.6	16.6	0.00	50.5	4.0	<0.001
	MPC-Perfect	3.15	49.9	21.3	21.4	0.01	49.3	4.1	<0.001
	Scenario-MPC	2.87	54.7	8.8	8.8	0.00	50.7	674	<0.001
	CARMA	2.88	54.3	9.8	9.8	0.00	50.3	31	–
	CARMA+conformal	2.89	54.3	9.9	10.0	0.00	50.6	31	1.000
	CARMA-Perfect	2.75	56.7	3.8	3.8	0.00	50.1	31	<0.001
	Oracle	2.66	58.2	0.0	0.0	0.00	51.8	–	–
$\rho = 0.7$	ASAP-Local	8.31	0.0	72.6	99.6	4.54	0.0	0.08	<0.001
	Greedy-Spatial	6.76	19.3	38.4	38.5	0.00	73.3	0.26	<0.001
	Forecast-Reserve	6.12	27.3	24.3	30.8	1.12	73.6	1.1	<0.001
	MPC	5.92	29.7	20.2	31.2	1.90	31.6	6.6	<0.001
	MPC-Protect	5.79	31.3	17.2	21.1	0.69	37.8	8.1	<0.001
	MPC+conformal	5.74	32.1	16.0	18.1	0.36	32.1	6.5	<0.001
	MPC-Perfect	5.85	30.5	18.8	30.5	1.96	31.4	6.3	<0.001
	Scenario-MPC	5.36	36.8	7.5	7.6	0.01	31.2	1453	<0.001
	CARMA	5.38	36.6	8.0	8.0	0.02	31.5	40	–
	CARMA+conformal	5.41	36.2	8.5	8.8	0.05	31.5	40	0.034
	CARMA-Perfect	5.21	38.7	4.0	4.2	0.04	31.0	40	<0.001
	Oracle	5.04	41.1	0.0	0.0	0.00	32.7	–	–

Table 4: Out-of-sample results in the temporal-only regime (no migration), 52 weeks of 2024 $\times$ 3 traces. Columns as in Table 3. Greedy-Spatial is omitted because without migration it coincides with ASAP-Local.

Load	Policy	Emis.	Red.	Gap	Adj.	Late	ms	p
$\rho = 0.3$	ASAP-Local	3.71	0.0	22.3	22.3	0.00	0.03	<0.001
	Forecast-Reserve	3.27	12.3	7.1	8.6	0.38	0.33	<0.001
	MPC	3.23	13.4	5.6	5.6	0.00	1.3	1.000
	MPC-Protect	3.30	11.6	7.8	7.8	0.00	1.6	<0.001
	MPC+conformal	3.24	13.3	5.7	5.7	0.00	1.3	1.000
	MPC-Perfect	3.09	17.8	0.2	0.3	0.01	1.4	<0.001
	CARMA	3.23	13.4	5.6	5.6	0.00	6.9	–
	CARMA+conformal	3.23	13.5	5.6	5.6	0.01	7.2	1.000
	CARMA-Perfect	3.08	17.8	0.2	0.2	0.01	7.3	<0.001
	Oracle	3.08	18.0	0.0	0.0	0.00	–	–
$\rho = 0.5$	ASAP-Local	6.11	0.0	18.0	18.2	0.05	0.05	<0.001
	Forecast-Reserve	5.54	9.5	6.6	18.3	3.03	0.66	<0.001
	MPC	5.47	10.9	5.0	6.8	0.52	2.5	<0.001
	MPC-Protect	5.58	9.1	7.1	8.7	0.48	2.9	<0.001
	MPC+conformal	5.48	10.7	5.3	6.9	0.47	2.5	<0.001
	MPC-Perfect	5.29	14.0	1.3	4.2	0.78	2.6	<0.001
	CARMA	5.46	11.0	4.8	6.0	0.36	9.7	–
	CARMA+conformal	5.48	10.7	5.3	7.2	0.54	9.8	0.006
	CARMA-Perfect	5.28	14.3	0.9	2.9	0.54	10	<0.001
	Oracle	5.23	15.1	0.0	0.0	0.01	–	–
$\rho = 0.7$	ASAP-Local	8.31	0.0	9.6	22.5	4.54	0.07	<0.001
	Forecast-Reserve	7.93	4.5	4.6	30.7	8.15	1.2	<0.001
	MPC	7.88	5.2	3.8	42.1	11.80	4.5	0.006
	MPC-Protect	8.00	3.8	5.4	42.8	11.54	5.1	<0.001
	MPC+conformal	7.90	5.0	4.1	42.0	11.54	4.5	<0.001
	MPC-Perfect	7.77	6.6	2.3	50.0	14.03	4.6	<0.001
	CARMA	7.87	5.3	3.7	32.9	9.26	14	–
	CARMA+conformal	7.90	4.9	4.2	40.6	11.09	14	<0.001
	CARMA-Perfect	7.75	6.9	1.9	41.9	11.85	14	<0.001
	Oracle	7.62	8.7	0.0	0.0	0.76	–	–

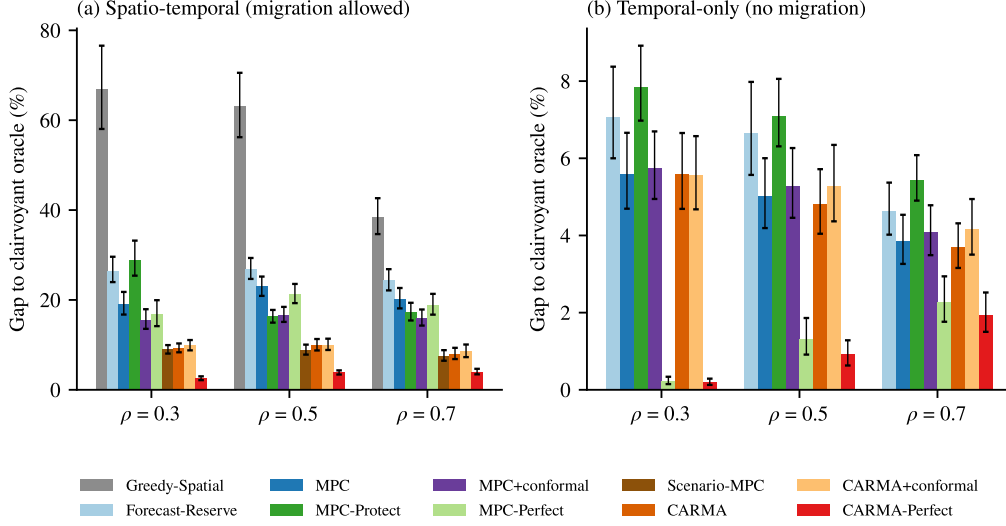


Figure 5: Mean gap to the clairvoyant oracle over the 52 test weeks of 2024 (week means of three traces) with 95% confidence intervals. ASAP-Local is omitted (gap 72–280% with migration, 9–22% without).

0.7 and MPC-Protect at $\rho = 0.5$ (16.3%); at $\rho = 0.3$, where clean capacity is rarely scarce, the protection level wastes it and MPC-Protect is worse than MPC (28.8%). CARMA reduces emissions by 69.3%, 54.3% and 36.6% and has gaps of 9.3%, 9.8% and 8.0%. Relative to MPC this is a further emission reduction of 7.7%, 9.7% and 9.1%. Its advantage over the best of these baselines at each load is 6.3 (95% block-bootstrap CI 4.5–8.2), 6.5 (5.2–7.4) and 8.0 (6.9–9.0) percentage points of gap, and CARMA emits less than MPC in 51, 51 and 52 of the 52 week-means. Scenario-MPC, which uses the same demand information in a two-stage stochastic programme, does slightly better than CARMA: its gaps are 9.0%, 8.8% and 7.5%, lower by 0.3 (0.2–0.5), 1.0 (0.9–1.3) and 0.4 (0.3–0.6) points, and CARMA is better in only 19, 5 and 8 weeks. Scenario-MPC, however, needs 250–1,450 ms per decision against 21–40 ms for CARMA, 12–37 times longer, and solves a general LP instead of a network flow. CARMA thus captures most of the value of demand information, 10–13 points of gap against MPC, and gives up 0.3–1.0 points against a stochastic programme with the same information. Block lengths of two and eight weeks give the same conclusions.

CARMA also finishes more work on time under heavy load. At $\rho = 0.7$,

0.02% of its work is late (Scenario-MPC 0.02%), against 1.9% for MPC, 0.7% for MPC-Protect, 1.1% for Forecast-Reserve and 4.5% for ASAP-Local. The service-adjusted gap therefore widens the difference: 8.0% for CARMA against 18.1–31.2% for the forecast-driven baselines without demand information. CARMA migrates about the same share of work as MPC (32–70%), whereas MPC-Protect migrates more (38–81%). CARMA’s advantage thus comes from better timing of the use of scarce clean capacity, not from moving more data. For the fleet of 1,700 batch servers at $\rho = 0.5$, CARMA emits 0.31 tCO₂ per week less than MPC, about 16 tCO₂ per year, and 3.22 tCO₂ per week less than carbon-agnostic operation. These absolute figures scale with fleet size. Two test weeks contain gap-filled hours (Section 3.5); excluding them changes CARMA’s gaps by at most 0.2 points and no ranking.

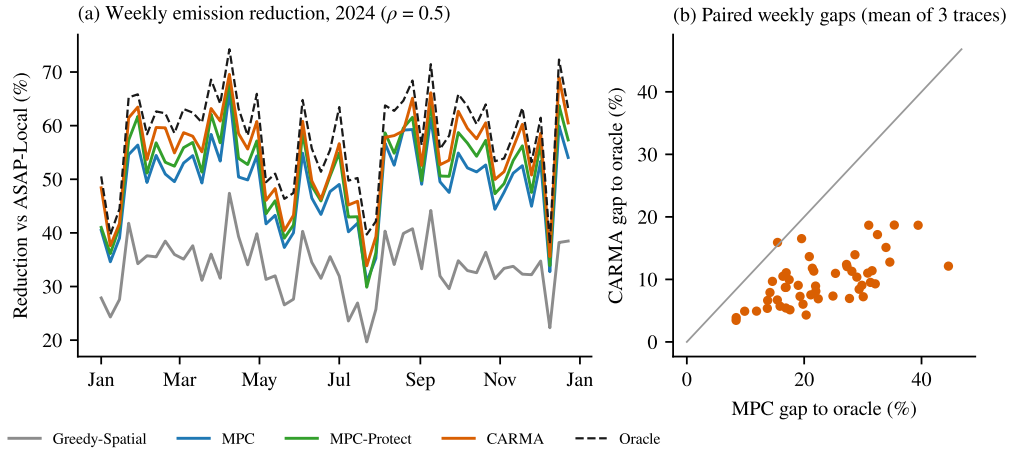


Figure 6: Weekly results in 2024 (spatio-temporal regime, $\rho = 0.5$, means of three traces). (a) Emission reduction relative to ASAP-Local. (b) Paired weekly gaps to the oracle; points below the diagonal are weeks in which CARMA outperforms MPC.

Value of information. The clairvoyant variants separate two sources of loss. With perfect intensity forecasts, the myopic MPC-Perfect still has gaps of 16.7%, 21.3% and 18.8%. Perfect carbon foresight is thus worth only 1.4–2.4 percentage points to a myopic planner, whereas anticipating demand is worth 9.8–13.2 points (MPC versus CARMA). With both, CARMA-Perfect comes within 2.5–4.0% of the oracle. CARMA-Perfect has exact carbon forecasts but still uses the average demand profile, so it is not the exact-prediction case of Theorem 2. The two kinds of information are complementary. Once

786 demand is anticipated, forecast accuracy is worth 4.0–6.8 points (CARMA
787 versus CARMA-Perfect), about three to four times more than for the myopic
788 planner. The ranking is stable across seasons (Fig. 6a), although the level of
789 achievable reductions varies from week to week with wind conditions.

790 *Temporal-only regime.* Without migration (Table 4), each job can only move
791 in time at its origin. Achievable reductions are much smaller: the oracle
792 saves 18.0%, 15.1% and 8.7%. MPC and CARMA are within 3.7–5.6% of
793 the oracle. They are statistically indistinguishable at $\rho = 0.3$ (difference 0.01
794 points, CI -0.01 to 0.02). CARMA is slightly better at higher loads, by 0.20
795 (0.13–0.30) and 0.15 (0.07–0.25) points. Both are better than Forecast-Reserve
796 and MPC-Protect. Anticipation brings little benefit here, because each site’s
797 capacity serves only its own jobs, and flexible and urgent work rarely compete
798 for the same clean hours in a way that deferral cannot absorb. The remaining
799 loss is due almost entirely to forecast error: the Perfect variants close the
800 gap to 0.2–2.3%. Heavy load exposes a trade-off that the emission metric
801 hides. At $\rho = 0.7$, the capacity at the origin sites is itself insufficient, and
802 every carbon-aware policy delays work: 9.3% of CARMA’s work and 11.8%
803 of MPC’s is late, against 4.5% for ASAP-Local. In this cell the emission
804 gap is not an optimality gap, because the oracle meets deadlines. Under the
805 service-adjusted metric, carbon-agnostic operation (22.5%) is preferable to
806 MPC (42.1%) and CARMA (32.9%). Without migration and near saturation,
807 temporal shifting should therefore be restricted or combined with explicit
808 service constraints.

809 *Computation.* A CARMA decision takes 7–40 ms on one CPU core, against
810 1–7 ms for MPC; the difference comes from the ghost jobs, which enlarge
811 the planning network. Scenario-MPC needs 0.25–1.5 s. Both are fast enough
812 for hourly decisions in this fleet, but the cost of the stochastic programme
813 grows with the number of scenarios and jobs, while CARMA remains a single
814 network flow. A clairvoyant weekly oracle with up to about 2,800 jobs is
815 solved in under one second by the flow solver.

816 5.4. Trace-derived workload

817 Table 5 reports the experiment with the trace-derived workload. Arrivals
818 and job sizes come from the Alibaba trace, and the demand information
819 of CARMA and Scenario-MPC comes from different days, so the profile
820 underestimates demand by 28% and cannot anticipate bursts. The evaluation

821 covers the four trace days paired with 52 carbon windows of 2024, with three
 822 draws of origins and deadlines each, and 3,072 evaluated job batches (65,300
 823 server-hours) per window. Real, bursty demand makes all policies worse than
 824 in the synthetic experiment, but the ranking is unchanged. CARMA has a gap
 825 of 21.2%, against 27.5% for MPC-Protect, the best baseline without demand
 826 information, and 36.8% for MPC. Carbon foresight again matters less than
 827 demand anticipation: MPC-Perfect reaches only 34.1%. CARMA’s advantage
 828 over MPC-Protect is 6.3 percentage points (95% CI 4.6–8.3; CARMA better
 829 in 42 of 52 carbon windows, $p < 0.001$). Scenario-MPC is again slightly
 830 better than CARMA, by 1.4 points (0.6–1.8; gap 19.7%), at 1.2 s per decision
 831 against 48 ms. The bias of the learned profile costs 3.0 points (CI 1.1–4.0):
 832 CARMA with a profile estimated in-sample from the evaluation days reaches
 833 18.2%. With the biased profile but perfect carbon forecasts, the gap would
 834 fall by 3.8 points (CARMA-Perfect, 17.4%). The remaining distance to the
 835 oracle reflects bursts that no profile anticipates. CARMA also keeps late work
 836 low (0.09%, against 0.66% for MPC).

837 Batching pools the widths of jobs with different work-to-width ratios
 838 and therefore relaxes the problem. With finer batches that also separate
 839 jobs by run-length class (about twice as many batches, one draw per carbon
 840 window), the oracle emits 2.5% more, so batching makes the benchmark
 841 slightly optimistic. The ranking is unchanged: gaps are 20.0% for CARMA,
 842 26.2% for MPC-Protect and 34.3% for MPC, and CARMA’s advantage over
 843 MPC-Protect is 6.3 points (4.8–7.8). The 156 episodes share one demand
 844 chronology, so the trace result is evidence about robustness to carbon condi-
 845 tions and to a biased profile for this workload, not a sample of independent
 846 demand weeks.

Table 5: Trace-derived workload (Alibaba 2018 cluster trace, days 5–8; demand information from days 1–4), paired with 52 carbon windows of 2024×3 draws of origins and deadlines; spatio-temporal regime, $\rho = 0.5$. Emissions are per four-day window. Columns as in Table 3; the service-adjusted gap is omitted; p values are Holm-adjusted within this table.

Policy	Emis.	Red.	Gap	Late	Mig.	ms	p
ASAP-Local	3.67	0.0	198.8	0.00	0.0	0.06	<0.001
Greedy-Spatial	2.48	33.4	94.2	0.00	75.0	0.19	<0.001
Forecast-Reserve	1.92	49.4	43.4	0.82	77.5	1.8	<0.001
MPC	1.84	51.7	36.8	0.66	52.9	16	<0.001
MPC-Protect	1.73	54.6	27.5	0.06	63.9	16	<0.001
MPC-Perfect	1.80	52.7	34.1	0.63	52.3	15	<0.001
Scenario-MPC	1.65	57.1	19.7	0.07	53.3	1160	<0.001
CARMA	1.66	56.7	21.2	0.09	53.4	48	–
CARMA-Perfect	1.61	58.1	17.4	0.15	51.9	49	<0.001
Oracle	1.40	63.8	0.0	0.00	56.5	–	–
CARMA (in-sample profile)	1.62	57.6	18.2	0.00	52.2	48	<0.001

847 5.5. Sensitivity analysis

848 Table 6 varies one factor at a time around the base case: spatio-temporal
849 regime, $\rho = 0.5$, $v = 0.3$, $\eta = 0.02$ kWh/GB, capacities 300/400/400/600.
850 Each setting uses 26 alternating test weeks of 2024 with two fresh workload
851 traces per week. All policies keep their base-case hyperparameters.

852 *Workload and network.* CARMA has the smallest gap of the online policies
853 compared here in every setting (Scenario-MPC is omitted from the sensitivity
854 analysis because of its computational cost). Except under marginal accounting
855 (discussed below), it emits less than MPC in all 26 weeks of each setting.
856 The benefit of anticipation grows with the share of urgent work, because
857 urgent arrivals are precisely the demand that myopic planners crowd out.
858 With $v = 0.5$, CARMA’s gap is 6.7% against 15.3% for MPC-Protect and
859 21.5% for MPC. With $v = 0.1$, it is 12.0% against 16.4% and 23.3%. The
860 ranking is unchanged over a twelve-fold range of transmission energy intensity
861 ($\eta = 0.005$ – 0.06 kWh/GB), and the migrated share of CARMA’s work stays
862 at 49–51%.

863 *Capacity layout.* Arrival rates are rescaled with the fleet’s capacity, so the
864 offered load stays at $\rho = 0.5$ of the new capacity. The advantage does not
865 depend on the chosen layout. When the clean South Scotland site is halved
866 (150 servers) or doubled (600), or London’s capacity is doubled, CARMA’s

Table 6: Sensitivity analysis (spatio-temporal regime; 26 alternating weeks of 2024×2 traces per setting). Oracle red.: emission reduction of the oracle relative to ASAP-Local (%). Gap columns: excess over the oracle (%). Wins: weeks in which CARMA emits less than MPC. Late: share of late work (%), MPC / CARMA. ^aBusy servers draw 0.2 kWh above an idle draw of 0.2 kWh per server-hour that every batch server incurs regardless of the schedule. ^bSchedules are made with average intensities, but emissions are accounted with the stylised marginal emission factor of Section 5.5, and the oracle optimises marginal emissions.

Setting	Oracle		Gap to oracle (%)					Wins vs		Late (%)	
	red.	red.	Greedy	Reserve	MPC	Protect	CARMA	MPC	MPC/CARMA	MPC	MPC/CARMA
Urgent share $v = 0.1$	62.0	62.0	79.2	28.5	23.3	16.4	12.0	26/26	0.00 / 0.00	26/26	0.00 / 0.00
Urgent share $v = 0.5$	56.4	56.4	53.8	25.5	21.5	15.3	6.7	26/26	0.09 / 0.00	26/26	0.09 / 0.00
Network energy $\eta = 0.005$ kWh/GB	62.3	62.3	70.1	28.7	27.3	18.0	10.9	26/26	0.01 / 0.00	26/26	0.01 / 0.00
Network energy $\eta = 0.06$ kWh/GB	53.9	53.9	65.5	28.8	19.4	14.9	8.4	26/26	0.00 / 0.00	26/26	0.00 / 0.00
Load $\rho = 0.4$, profile learned at 0.5	67.3	67.3	75.9	29.0	23.3	15.9	12.2	26/26	0.00 / 0.00	26/26	0.00 / 0.00
Load $\rho = 0.6$, profile learned at 0.5	51.3	51.3	56.0	27.8	24.1	19.0	8.6	26/26	0.31 / 0.00	26/26	0.31 / 0.00
Clean-site capacity halved (150)	52.4	52.4	58.2	25.4	21.7	16.9	9.5	26/26	0.02 / 0.00	26/26	0.02 / 0.00
Clean-site capacity doubled (600)	69.2	69.2	85.1	32.7	28.6	17.5	9.3	26/26	0.01 / 0.00	26/26	0.01 / 0.00
London capacity doubled (1200)	47.6	47.6	53.1	27.9	22.7	18.9	9.5	26/26	0.00 / 0.00	26/26	0.00 / 0.00
Always-on servers (idle 0.2 kWh) ^a	18.5	18.5	10.9	4.8	3.6	2.6	1.5	26/26	0.00 / 0.00	26/26	0.00 / 0.00
Marginal-emission accounting ^b	13.8	13.8	12.2	10.7	8.6	9.2	7.9	18/26	0.01 / 0.00	18/26	0.01 / 0.00

gap stays at 9.3–9.5%. The gaps of MPC (22–29%) and MPC-Protect (16.9–18.9%) stay about two to three times larger. More clean capacity raises the oracle’s reduction (from 52.4% to 69.2%), but a myopic planner still misallocates it.

Misspecified demand profile. CARMA’s ghost jobs come from a profile learned at $\rho = 0.5$. When the actual load is 20% higher ($\rho = 0.6$), so that CARMA under-anticipates, its gap is 8.6% against 19.0% for MPC-Protect and 24.1% for MPC. When the actual load is 20% lower ($\rho = 0.4$), CARMA over-anticipates and reserves more clean capacity than needed. Its gap rises to 12.2%, but it still beats MPC-Protect by 3.7 points (15.9%) and MPC by 11.2 points. The trace experiment of Section 5.4 adds a real profile that underestimates demand by 28%. Because ghost work carries a lower penalty than real work, the planner never plans real work late in order to serve ghost demand. Without a backstop, however, a misprediction can still delay real work once capacity is exhausted (Section 4.2); the late-work column of Table 6 tracks this.

Always-on servers. If every batch server draws idle power whether or not it is busy, a large part of the emissions no longer depends on the schedule. The achievable reduction falls to 18.5% for the oracle and 17.3% for CARMA. The ranking is unchanged (gaps of 1.5% for CARMA, 2.6% for MPC-Protect and 3.6% for MPC). Carbon-aware scheduling is therefore most valuable when it is combined with right-sizing, so that deferred work actually reduces the number of powered servers [18].

Marginal emissions. All sites are in one GB market, where gas-fired plant is usually the marginal generator [7, 46]. Moving load between regions then mainly changes *which* region is assigned the average mix, and it abates emissions physically only when low-carbon output would otherwise be curtailed. We therefore account emissions with a stylised marginal emission factor (MEF). The MEF is 394 gCO₂/kWh (combined-cycle gas) in every hour except two cases where it is zero. For South Scotland it is zero when Scotland is nearly carbon-free (below 25 g) while GB-wide intensity exceeds 100 g, a proxy for export-constrained hours; this holds in 27% of hours. For all sites it is zero when GB-wide intensity is below 50 g (4% of hours). Under this accounting the picture changes substantially. The oracle cuts marginal emissions by only 13.8%, and CARMA by 7.5%. The forecast-driven policies are within 7.9–10.7% of the marginal oracle. CARMA is still the best of them,

but its advantage over MPC shrinks to 0.6 points (block-bootstrap CI 0.1–0.9; $p = 0.003$ after Holm correction; CARMA better in 18 of 26 weeks). The schedules were optimised for average intensity, not for the MEF, and the MEF itself is a proxy with fixed thresholds that has not been validated against dispatch or curtailment data. Under this proxy, most of the attributional reductions reported above would be reallocations within the GB system. The physical benefit depends on whether shifting absorbs otherwise-curtailed low-carbon output or changes the marginal generator between hours, and a scheduler should be driven by a validated marginal signal when that is the goal.

5.6. Other grids and continents

To test whether the findings depend on the British grid, we repeat the experiment on the four further fleets of Table 2: Europe, the United States, Brazil and a fleet with one site on each of four continents. The last fleet spans time zones from UTC−8 to UTC+12, so jobs submitted during one region’s working day can run in another region’s night. For every fleet we retrain the forecaster and tune CARMA, MPC-Protect and the conformal view of MPC on the fleet’s own validation weeks (Section 3.5), so no method keeps a setting tuned on Great Britain. Table 7 and Fig. 7 summarise the results at $\rho = 0.5$, and Appendix D gives all loads.

Table 7: Results on five grids ($\rho = 0.5$, migration allowed; 52 test weeks of 2024). Red.: emission reduction of the oracle relative to ASAP-Local (%). Columns 3–6: gap to the oracle (%). Best-base.: gap of the best of Greedy-Spatial, Forecast-Reserve, MPC, MPC-Protect and MPC+conformal minus CARMA’s gap (points). Wins: weeks in which CARMA emits less than MPC. Scenario-MPC gap / CARMA gap on the 26 alternating weeks on which Scenario-MPC was run (one trace). Great Britain: main experiment (three traces per week); other fleets: two traces per week.

Fleet	Oracle red.	MPC	MPC-Prot.	MPC+conf.	CARMA	Scen.-MPC / CARMA	Best-base.	Wins	CARMA-Perfect
Great Britain	58.2	23.0	16.3	16.6	9.8	9.0 / 10.1	6.5	51/52	3.8
EU	63.0	19.7	13.0	17.5	5.8	5.7 / 5.6	7.3	52/52	5.3
US	34.6	7.0	7.7	6.7	4.3	3.9 / 4.3	2.4	52/52	3.7
BR	37.5	6.6	6.0	4.4	2.7	2.9 / 2.7	1.7	52/52	2.3
GLOBAL	56.0	9.2	6.8	7.2	4.0	4.5 / 4.3	2.8	52/52	3.4

CARMA has the smallest gap of all implementable policies without a stochastic programme on every grid and at every load. At $\rho = 0.5$ its gaps are 5.8% in Europe, 4.3% in the United States, 2.7% in Brazil and 4.0% for

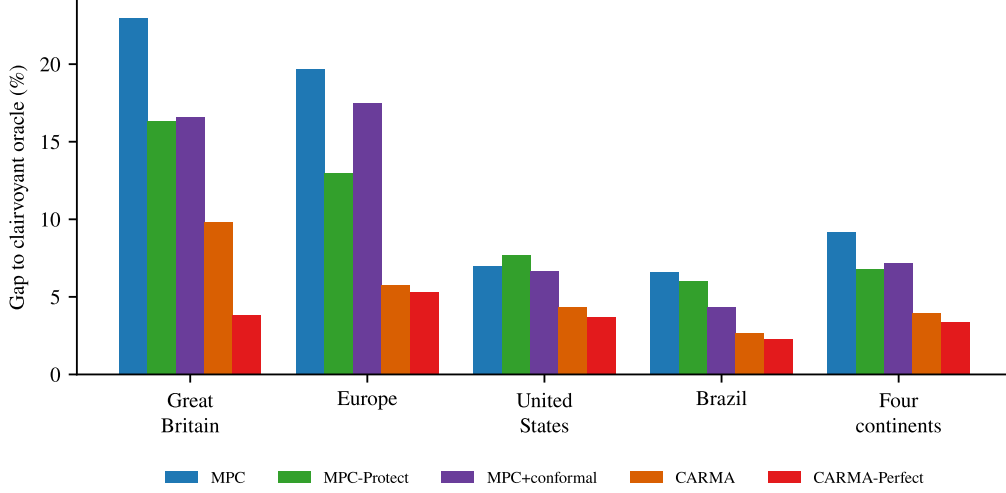


Figure 7: Gap to the clairvoyant oracle on five grids ($\rho = 0.5$, migration allowed).

926 the four-continent fleet, against 13.0%, 6.7%, 4.4% and 6.8% for the best
 927 baseline without demand information. CARMA's advantage is 7.3 (95%
 928 block-bootstrap CI 6.2–8.7), 2.4 (2.0–2.9), 1.7 (1.2–2.3) and 2.8 (2.4–3.3)
 929 points, and it emits less than MPC in all 52 weeks on each grid. Which
 930 baseline is best varies. MPC-Protect is the best baseline in Europe and for
 931 the four-continent fleet and MPC+conformal in the United States and Brazil,
 932 whereas at $\rho = 0.3$ MPC-Protect is worse than plain MPC on all four grids.
 933 Over all 15 fleet-load cells the advantage over the best baseline ranges from 1.3
 934 to 7.3 points. It is smaller than in Great Britain (6.3–8.0 points) in the United
 935 States and Brazil, and the gap of myopic MPC is correspondingly smaller
 936 (7.0% and 6.6% at $\rho = 0.5$, against 23.0% in Great Britain). A possible reason
 937 is that the sites of these two fleets are closer to each other relative to their
 938 mean intensity: the median relative spread between the cleanest and the
 939 dirtiest site, $(\max - \min)/\text{mean}$ in 2024, is 1.25 in the United States and 1.27
 940 in Brazil, against 1.71 in Great Britain, 1.80 for the four-continent fleet and
 941 2.06 in Europe, where the nuclear-based French and the coal-based Polish
 942 grids are far apart. The advantage broadly follows this spread, but five grids
 943 cannot establish a relation. Perfect carbon forecasts would reduce CARMA's
 944 gap by a further 0.4–0.6 points at $\rho = 0.5$, so the remaining distance to the
 945 oracle is mostly due to demand uncertainty. CARMA's late work stays below
 946 0.03% of work at every load.

947 The emission reductions follow the same pattern. At $\rho = 0.5$ CARMA
 948 reduces emissions relative to carbon-agnostic operation by 60.9% in Europe,
 949 31.8% in the United States, 35.9% in Brazil and 54.2% for the four-continent
 950 fleet, compared with 63.0%, 34.6%, 37.5% and 56.0% for the oracle. These
 951 are attributional reductions under production-based intensities and inherit
 952 the caveats of the marginal analysis above.

953 *Scenario-MPC.* The stochastic lookahead is level with CARMA on three of
 954 the new grids. Its gap is 0.09 points above CARMA’s in Europe (CI -0.24
 955 to 0.42), 0.15 points above it in Brazil (0.01 – 0.34) and 0.20 points above
 956 it for the four-continent fleet (-0.05 to 0.55). It is better by 0.44 points
 957 in the United States (0.25 – 0.69), and by 1.0 points in Great Britain on the
 958 same subset of weeks. It needs 0.9 s per decision against 33 ms for CARMA,
 959 26–27 times longer. The overall picture from the British experiment therefore
 960 holds on other grids: CARMA captures nearly all of the value of demand
 961 information at a small fraction of the computing cost of a scenario-based
 962 programme.

963 *Reversed capacity layout.* In all fleets the cleanest site has the least capacity
 964 and the smallest share of job origins, as in Great Britain. To check that
 965 the result does not depend on this layout, we reverse both assignments, so
 966 that the cleanest site is the largest. CARMA’s gap is then 6.7%, 6.0%, 2.6%
 967 and 3.6% in Europe, the United States, Brazil and the four-continent fleet,
 968 against 18.9%, 12.4%, 4.6% and 6.6% for the best baseline without demand
 969 information (26 alternating weeks, $\rho = 0.5$; advantages of 12.1, 6.4, 2.0 and
 970 3.0 points). The oracle’s reductions are 61.7%, 40.7%, 28.7% and 46.3% in
 971 this layout, and CARMA’s 59.2%, 37.2%, 26.9% and 44.4%. The ranking of
 972 the policies is unchanged.

973 *Forecasts.* The forecaster is more accurate than persistence on every grid
 974 (mean absolute error of 30.1 against 52.4 gCO₂/kWh in Europe, 19.8 against
 975 34.8 in the United States, 7.3 against 12.3 in Brazil and 10.6 against 21.7 for
 976 the four-continent fleet), and the adaptively recalibrated upper bounds cover
 977 within 0.012 of the target at every level.

978 5.7. Robustness to prediction errors

979 We test how CARMA degrades as its predictions deteriorate on every
 980 fourth test week of 2024 (13 weeks, $\rho = 0.5$, migration allowed, fresh workload
 981 traces). Controlled errors are injected into the two kinds of prediction (Fig. 8).

982 These experiments do not satisfy the assumptions of the bound in Theorem 2,
 983 so they probe the degradation profile empirically.

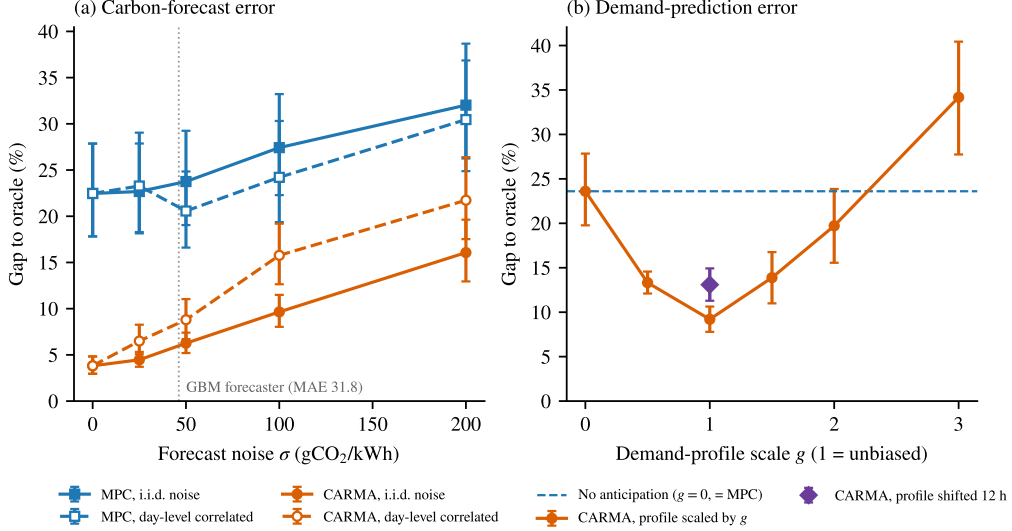


Figure 8: Degradation under controlled prediction errors (13 test weeks of 2024, $\rho = 0.5$, migration allowed; means with 95% confidence intervals). (a) Carbon-intensity forecasts replaced by the realised values plus Gaussian noise of standard deviation σ . The noise is either independent across issue times and horizons (solid) or shared by all forecasts of the same site and target day (dashed, open markers). (b) Demand profile scaled by g , or shifted by 12 hours so that arrivals are predicted at the wrong time of day.

984 *Carbon-forecast error.* With exact forecasts, CARMA’s gap is 3.8% and
 985 MPC’s 22.5%. Independent noise is comparatively benign: CARMA’s gap
 986 rises smoothly to 4.5%, 6.3%, 9.7% and 16.1% at $\sigma = 25, 50, 100$ and
 987 200 gCO₂/kWh. Noise that persists over a whole target day, as when a
 988 weather front is missed, is more damaging. The gaps are then 6.5%, 8.8%,
 989 15.8% and 21.7%, because the planner cannot average out the error by re-
 990 planning. MPC deteriorates from 22.5% to 30.5–32.0% at the largest noise
 991 level. At every noise level and type, CARMA remains better than MPC with
 992 the same forecasts. Only with the most persistent large errors ($\sigma = 200$) does
 993 it approach the level of MPC with *perfect* forecasts. Fig. 8a places the GBM
 994 forecaster at the noise level with the same realised MAE (31.8 gCO₂/kWh,
 995 $\sigma \approx 46$). With the GBM forecasts, CARMA’s gap on the same weeks is
 996 9.9%, larger than with either noise model at that error size (about 6% and

997 8.5%). The forecaster’s errors, which are both serially and cross-sectionally
 998 correlated and concentrated in volatile hours, are therefore more damaging
 999 than either stylised noise model, and the degradation should not be read as
 1000 merely i.i.d.

1001 *Demand-prediction error.* Scaling the learned demand profile by g produces a
 1002 V-shaped response with its minimum at the unbiased profile ($g = 1$, gap 9.2%).
 1003 Under-prediction by half ($g = 0.5$) and over-prediction by half ($g = 1.5$) cost
 1004 4.1 and 4.7 percentage points. Predicting arrivals at the wrong time of day
 1005 (a 12-hour shift) costs 3.9 points. Removing anticipation ($g = 0$) returns
 1006 the MPC level (23.6%). Gross over-prediction ($g = 3$, three times the true
 1007 demand) is worse than no anticipation (34.2%), because CARMA then holds
 1008 clean capacity for work that never arrives. A 12-hour shift keeps most of the
 1009 benefit, which indicates that much of the value comes from anticipating the
 1010 *volume* of demand rather than its exact timing. Late work was negligible
 1011 in all 325 runs (at most 0.01% of work). The largest ratio of CARMA’s
 1012 emissions to the oracle’s over all runs was 1.58.

1013 5.8. Fair carbon attribution

1014 We evaluate Theorem 3 on the 52 test weeks at $\rho = 0.5$, with the four
 1015 sites as players. For every week and each of the 15 coalitions we compute
 1016 the hindsight value $c(C)$, and we also run CARMA with only the coalition’s
 1017 sites and jobs. Each game is one 216-hour arrival episode per week: the
 1018 evaluated week plus 24 hours of arrivals before and after it. Pooling pays: the
 1019 hindsight optimum of the grand coalition has a penalised cost of 3.32 tCO₂
 1020 per episode, against 6.69 for the four sites operating separately; the emissions
 1021 alone are 3.32 and 6.58 tCO₂. With CARMA the figures are 3.58 and 6.87.
 1022 The question is how to share the reduction.

1023 We compare six attributions. The *origin-based* rule charges each site
 1024 the emissions of its own jobs wherever they run. The *host-based* rule, the
 1025 analogue of location-based Scope 2 accounting, charges each site the emissions
 1026 of the work it hosts. The *work-proportional* rule splits the total in proportion
 1027 to submitted work, and the *stand-alone proportional* rule in proportion to
 1028 stand-alone emissions. The *Shapley value* [67] averages marginal contributions
 1029 over all orders of joining. The *dual* rule is Eq. (10). Because $c(C)$ includes
 1030 the lateness penalty whenever a coalition cannot serve all of its work, we
 1031 also evaluate an emissions-only game, in which $c(C)$ is the emissions of the
 1032 hindsight schedule. Stand-alone coalitions leave work unserved in 8 of the 52

1033 episodes (15%), up to 256 server-hours in one episode. In that game the dual
1034 attribution is rescaled to the emissions total and is no longer covered by the
1035 theorem. For CARMA’s realised emissions, the dual attribution is applied to
1036 the hindsight value and the online excess over the hindsight optimum is split
1037 in proportion to work. A check is violated when a coalition is charged more
1038 than it would emit on its own.

Table 8: Coalitional stability of carbon attributions (52 test weeks of 2024, $\rho = 0.5$; one 216-hour arrival episode per week). Checks: share of the 52×14 (episode, proper coalition) pairs in which the coalition is charged more than its stand-alone value. Weeks: share of episodes with at least one violation. IR: share of (episode, site) pairs in which a site is charged more than its stand-alone value. Neg.: share of (episode, site) pairs with a net credit. Max. exc.: largest violation (tCO₂). Panel (a) uses the penalised values $c(C)$ of Theorem 3, panel (b) emissions only, panel (c) CARMA’s realised emissions.

Attribution	Checks (%)	Weeks (%)	IR (%)	Neg. (%)	Max. exc.
<i>(a) Hindsight, penalised cost</i>					
Origin-based	46.4	100.0	49.0	0.0	1.25
Host-based	49.2	100.0	50.0	0.0	4.30
Work-proportional	44.5	100.0	45.2	0.0	1.43
Stand-alone proportional	22.4	100.0	0.0	0.0	1.05
Shapley	3.6	44.2	0.0	41.3	0.20
Dual	0.0	0.0	0.0	38.0	–
<i>(b) Hindsight, emissions only</i>					
Shapley	3.2	42.3	0.0	41.3	0.13
Rescaled dual	0.0	0.0	0.0	38.0	–
<i>(c) CARMA, realised emissions</i>					
Origin-based	45.3	100.0	45.7	0.0	1.20
Host-based	48.6	100.0	50.0	0.0	4.04
Work-proportional	44.5	100.0	45.2	0.0	1.48
Stand-alone proportional	21.7	100.0	0.0	0.0	1.00
Shapley	3.6	50.0	0.0	41.3	0.12
Adjusted dual (heuristic)	0.5	7.7	0.0	35.6	0.03

1039 Table 8 and Fig. 9 report the results. In the hindsight game the dual
1040 attribution satisfies all 728 coalition checks, as Theorem 3 guarantees. The
1041 dual prices are not always unique, but over all optimal duals (checked in 9
1042 episodes) the attribution of a site varies by at most 0.011 tCO₂, so the choice
1043 of dual does not matter in practice. The four accounting rules violate 22–49%
1044 of the checks and fail in every episode. The origin-based rule is individually
1045 irrational for South Scotland in all 52 episodes and for North West England

1046 in 50: their own jobs are displaced to dirtier capacity by imported work, so
 1047 they are charged more than they would emit on their own. The host-based
 1048 rule fails for the same two sites in every episode, because they are charged for
 1049 the imported work they host; North West England, for example, is charged
 1050 1.5 tCO₂ per episode against 0.6 on its own. Scaling stand-alone emissions
 1051 down is individually rational by construction, but it still leaves 22% of the
 1052 coalition checks violated. The Shapley value comes close: it is individually
 1053 rational and violates 3.6% of the checks (44% of episodes), with a largest
 1054 excess of 0.20 tCO₂. Its site charges are similar to the dual ones. Both credit
 1055 the clean sites for the capacity they contribute. Under the dual rule South
 1056 Scotland receives a net credit of 0.95 tCO₂ per episode (Shapley 0.80), North
 1057 West England is close to neutral, and London's charge (2.81 t) remains well
 1058 below its stand-alone value (3.91 t). Net credits arise in 38% of the (episode,
 1059 site) pairs. The Shapley value also needs all 2^n coalition values, whereas the
 1060 dual rule comes from a single solve. The emissions-only game gives almost the
 1061 same picture: the rescaled dual attribution passes every check, and Shapley
 1062 fails 3.2%.

1063 With CARMA's realised emissions, which is the practical case, the adjusted
 1064 dual attribution remains individually rational for every site in every episode.
 1065 It violates 4 of the 728 coalition checks (0.5%, in 4 of 52 episodes), all
 1066 for the coalition of South Scotland, North West England and London, and
 1067 the largest excess is 0.03 tCO₂. Shapley applied to the online values fails
 1068 3.6% of the checks, and the four accounting rules fail 22–49%. Stand-alone
 1069 coalitions under CARMA leave at most 3.7% of their work late, and unfinished
 1070 work emits nothing, so the online checks, if anything, favour secession. The
 1071 online result is an empirical observation for this heuristic adjustment, which
 1072 Theorem 3 does not cover.

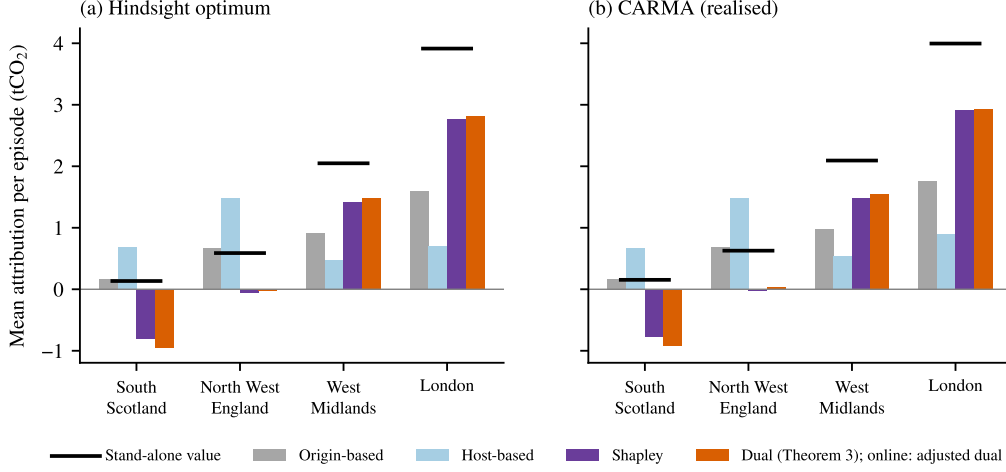


Figure 9: Mean weekly carbon attribution by site under four rules, compared with each site’s stand-alone value (black lines), 52 test weeks of 2024 at $\rho = 0.5$. (a) Hindsight optimum. (b) CARMA’s realised emissions. Negative values are net credits for contributing low-carbon capacity.

6. Discussion

6.1. Managerial implications

Protect scarce clean capacity. In multi-site carbon-aware scheduling, the largest operational loss does not come from imperfect carbon forecasts. It comes from spending low-carbon capacity on work that could have waited or run elsewhere. Proposition 2 and Theorem 1 show this in the worst case, and the experiments confirm it on average. With migration allowed, a myopic planner with perfect carbon foresight is still 17–21% above the optimum in the synthetic experiment and 34% above it on the trace-derived workload. Clean capacity at a low-carbon site should be treated as a scarce resource with an opportunity cost, as in revenue management. A simple protection level (MPC-Protect) captures part of this value. CARMA captures more, because its ghost jobs price the protection hour by hour from the operator’s own history. The result held when the profile underestimated demand by 28% and demand was bursty. A stochastic lookahead over sampled demand scenarios does slightly better in Great Britain and the United States (0.4–1.0 points of gap) and is level with CARMA elsewhere, at 12–37 times the decision time; when the planning network can be solved in seconds, it is a reasonable

1091 alternative. The value of demand information is not specific to one grid: the
1092 advantage of CARMA over the best baseline without it is 7.3 points in Europe
1093 and 1.7–2.8 points in the other three fleets, and it tends to be smaller where
1094 the sites are closer to each other relative to their mean intensity.

1095 *Invest in forecasts where they pay.* The value of better intensity forecasts
1096 depends on the rest of the policy. For a myopic multi-site planner, perfect
1097 forecasts are worth only 1.4–2.4 percentage points; once demand is anticipated
1098 they are worth 4–7 points; without migration they are worth almost the whole
1099 remaining gap. Persistent errors, such as a missed day-ahead wind front,
1100 are more costly than independent ones. Forecasting effort, for example the
1101 integration of numerical weather predictions [44, 43], should therefore target
1102 day-level biases, and in multi-site settings it should follow the fix of demand
1103 myopia.

1104 *Use risk premia with care.* The conformal cost views help a myopic planner
1105 but not CARMA. They work by making volatile sites and distant hours look
1106 dirtier, which crudely imitates anticipation. Once demand is anticipated,
1107 the same distortion gives up useful deferrals. Calibrated quantiles are better
1108 suited to reporting and to service-level decisions than to reshaping costs by
1109 default.

1110 *Emissions or service near saturation.* Without migration and at high load,
1111 every carbon-aware policy delayed work, and under a service-adjusted metric
1112 carbon-agnostic operation was preferable. Operators should combine temporal
1113 shifting with explicit service constraints, or restrict it, when local capacity is
1114 close to saturation.

1115 *Attributional versus physical savings.* The headline reductions are attribu-
1116 tional: they measure how much of the work runs where and when the *average*
1117 regional mix is clean. Within a single market such as Great Britain, moving
1118 load between regions mostly reassigns the average mix. Under a stylised
1119 marginal-emission proxy, the achievable reduction fell from 58% to 14% and
1120 CARMA’s advantage over MPC became small (0.6 points). Carbon-aware
1121 shifting is physically most valuable when it absorbs low-carbon output that
1122 would otherwise be curtailed, for example behind transmission constraints
1123 between Scotland and England, when temporal shifts change which plant is
1124 marginal, or when sites lie in different markets. Our marginal accounting is
1125 a stylised proxy, not an estimate of GB-wide abatement, so it indicates the

direction and rough size of this effect rather than its value. In such settings a marginal signal should drive the scheduler. The scheduling framework is unchanged: only the cost coefficients of the flow network change. Idle power works in the same direction. If servers stay powered, deferral saves only their dynamic energy, and carbon-aware scheduling should be paired with right-sizing [18].

Fair internal carbon accounting. Pooling halved the hindsight emissions of the four sites. However, origin-based accounting, which charges each site for its own jobs wherever they run, would have made South Scotland worse off than on its own in every episode. A pool of organisations or business units allocated this way is unstable, because its cleanest members have an incentive to leave. The dual attribution of Theorem 3 charges each site the carbon shadow price of its demand and credits it for the capacity it contributes. It is budget-balanced, no coalition would prefer to secede, and it is a by-product of the scheduling problem. It suits internal carbon pricing and the sharing of the benefits of carbon-aware scheduling among business units. It is not a substitute for external GHG Protocol reporting, where negative Scope 2 emissions do not exist [66]. The players must also be entities that could credibly schedule on their own, such as separate operators, business units with their own capacity, or tenants with reserved shares. Because credits reward contributed capacity, capacity declarations should be audited.

Deployment. CARMA needs a carbon-intensity feed (average or marginal), a demand profile computed from the operator’s own job logs and a network-flow solver. It decides within tens of milliseconds and returns a plan that is optimal for its model, which makes its decisions auditable. Absolute savings scale with the fleet: for the 1,700-server fleet simulated here, CARMA saves about 16 tCO₂ per year relative to myopic control and about 165 tCO₂ relative to carbon-agnostic operation, in attributional terms.

6.2. Limitations and future research

The study has several limitations. First, the synthetic workloads use a profile learned from the same generator, which favours anticipation. The trace experiment partly addresses this concern, but it covers only four evaluation days of one cluster, arrival times are first task starts, work is based on planned CPU requests, deadlines and origins are synthetic, and jobs are batched. Its 156 episodes vary carbon conditions but share one demand chronology. Second,

1161 the regional intensities published by NESO are modelled estimates, used here
 1162 as realised values, 52 hours of them are missing and filled by persistence, and
 1163 the marginal analysis relies on a stylised proxy with fixed thresholds, not
 1164 on a dispatch model or observed curtailment. Third, the facility parameters
 1165 (PUE, network energy, a 96-core server) are stylised, and embodied emissions
 1166 are outside the scope except as the floor that makes θ finite. Fourth, the
 1167 forecaster excludes weather inputs, so its day-ahead skill is limited. Fifth,
 1168 bag-of-tasks jobs with free preemption and migration are an idealisation: jobs
 1169 with state, data gravity, bandwidth limits or data-sovereignty constraints
 1170 would add restrictions that break the pure network structure. Sixth, the
 1171 guarantees of Theorem 2 assume a backstop resource, a positive emission
 1172 floor and an episode within the planning window, none of which holds in the
 1173 simulations. The theorem describes the mechanism and the experiments its
 1174 empirical profile. Its robustness term is shared by all on-time policies and is
 1175 far from the $\Omega(\sqrt{\theta})$ lower bound. Seventh, the core property is exact only for
 1176 the hindsight benchmark, and the online adjustment is a heuristic. Eighth,
 1177 Scenario-MPC was tuned and tested with a modest number of scenarios and
 1178 was not included in the sensitivity analysis because of its cost. Ninth, the
 1179 additional fleets use production-based lifecycle intensities computed from
 1180 open generation data, uniform PUE and a capacity layout that mirrors the
 1181 British one (tested against its reverse), and they cover Europe, the Americas
 1182 and Oceania only: no open hourly generation data could be retrieved for Asia
 1183 or Africa, and the New Zealand site is a single national grid.

1184 Several extensions follow naturally. The ghost-demand profile could be
 1185 replaced by probabilistic arrival forecasts, with the anticipation weight set
 1186 by a newsvendor-type argument, or by scenario-based planning that treats
 1187 demand and carbon uncertainty jointly [57]. Tightening robustness towards
 1188 the lower bound while retaining consistency, for example by protecting deferral
 1189 decisions with reservation prices [24, 26], is an open theoretical question. The
 1190 framework can also be driven by marginal emission signals, coupled with
 1191 right-sizing, on-site renewables and storage [21, 22] and demand-response
 1192 programmes [23], and evaluated across several electricity markets.

1193 7. Conclusions

1194 This paper studied carbon-aware scheduling of deferrable computing work-
 1195 loads across data-centre sites as a sustainable-operations problem. The offline
 1196 problem, including width limits, deadlines and the emissions of data transfer,

is a minimum-cost flow, which yields exact oracles (after quantisation) and fast online planning. Uncertainty about demand alone forces every online algorithm to a competitive ratio of order $\sqrt{\theta}$, even with perfect carbon forecasts, and myopic planning does far worse. CARMA augments the planning network with ghost jobs for expected arrivals. In an idealised setting with a backstop and episodes inside the planning window, it meets every deadline under any predictions, is optimal under exact predictions and degrades linearly with prediction error; a numerical check in such episodes confirmed this. The dual prices of the pooled problem yield a carbon attribution that is budget-balanced and in the core of the hindsight pooling game.

In 52 out-of-sample weeks of 2024 on British regional grid data, with migration allowed, CARMA cut attributional emissions relative to carbon-agnostic operation by 37–69%. It came within 8.0–9.8% of the clairvoyant optimum, against 15.5–16.3% for the best baseline without demand information, and emitted less than myopic MPC in 51 or 52 of the 52 weeks. A scenario-based stochastic lookahead with the same demand information was level with CARMA or at most 1 point better, but 12–37 times slower. On four further fleets in Europe, the United States, Brazil and across four continents, CARMA’s gap was 2.7–5.8% against 4.4–13.0% for the best baseline without demand information, in every case with all 52 weeks in its favour against myopic MPC. On a trace-derived workload, with a demand profile that underestimated demand by 28% and bursty arrivals, CARMA’s gap was 21.2% against 27.5% for the best baseline without demand information. Anticipating demand was worth far more than perfect carbon foresight, and correlated day-level forecast errors hurt more than independent ones. Three results qualify these gains. Without migration, reductions are small (9–18%), and near saturation carbon-aware shifting delays work enough that carbon-agnostic operation is preferable under a service-adjusted metric. Under a stylised marginal-emission proxy, the achievable reduction fell to 14% and CARMA’s advantage over MPC shrank to 0.6 points, so the physical value of shifting within one grid depends on absorbing otherwise-curtailed low-carbon output. With always-on servers, most emissions no longer depend on the schedule.

Origin-based and host-based carbon accounting violated coalitional stability in every test episode. The dual attribution never did in hindsight, and failed 4 of 728 checks, by at most 0.03 tCO₂, for the heuristic adjustment to CARMA’s realised emissions. Its charges were close to the Shapley value, which needs all coalition values, whereas the dual attribution needs one opti-

1235 misation. Carbon-aware computing is most effective when schedulers plan for
1236 the work that has not yet arrived, not only for the grid conditions that lie
1237 ahead, and when its benefits are measured and shared on a sound basis.

1238 **CRediT authorship contribution statement**

1239 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Data
1240 curation, Formal analysis, Investigation, Validation, Visualization, Writing –
1241 original draft, Writing – review & editing.

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1243 The author declares no known competing financial interests or personal
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1249 **Data availability**

1250 The carbon-intensity data are publicly available from the NESO Carbon
1251 Intensity API [45] under the CC BY 4.0 licence. The generation-mix data
1252 of the additional grids are publicly available from Energy-Charts, the EIA-
1253 930 bulk files, the ONS open-data portal and the New Zealand Electricity
1254 Authority. The data snapshot used in this study, all code, the workload
1255 generator with fixed random seeds, and all raw experiment outputs are
1256 available from the author’s repository ([https://github.com/vinhqdang/](https://github.com/vinhqdang/Sustainable-Operations-and-Computers)
1257 [Sustainable-Operations-and-Computers](https://github.com/vinhqdang/Sustainable-Operations-and-Computers)).

1258 **Appendix A. Proofs**

1259 *Atoms.* Splitting a job j into sub-jobs $j_1, \dots, j_k$ with work w_{j_i} and widths
1260 $m_{j_i} = m_j w_{j_i} / w_j$ does not change the optimal value of any planning problem.
1261 Given a solution for the merged job, the allocation $x_{j_i s}(t) = x_{j s}(t) w_{j_i} / w_j$
1262 satisfies every constraint of the split problem at the same cost. Conversely,
1263 summing the sub-job allocations gives a feasible allocation for the merged

1264 job, because $\sum_i m_{j_i} = m_j$. Splitting also preserves window-feasibility, since
 1265 $w_{j_i}/m_{j_i} = w_j/m_j$. We may therefore compare true and predicted job sets
 1266 atom by atom.

1267 *Appendix A.1. Proof of Theorem 1*

1268 Normalise $L = 1$, so $U = \theta$. Site A has one server with cost $p \in [1, \theta]$ in
 1269 hour 0 and cost 1 in hour 1. Site B has ample capacity at cost θ in both
 1270 hours. Transfer costs are zero. A flexible job ($w = m = 1$, window $\{0, 1\}$) is
 1271 released at $t = 0$. The adversary either releases nothing more (scenario 1) or
 1272 releases an urgent job ($w = m = 1$, window $\{1\}$) at $t = 1$ (scenario 2). All
 1273 costs are known in advance.

1274 Consider any deterministic online algorithm that may split work fraction-
 1275 ally, and let $x \in [0, 1]$ be the amount of flexible work it runs at $t = 0$. Running
 1276 at B in hour 0 costs $\theta \geq p$ and is dominated, so we may assume this work runs
 1277 at A . In scenario 1 the algorithm finishes the remaining $1 - x$ at A in hour 1,
 1278 so $\text{ALG}_1 = xp + (1 - x)$, while $\text{OPT}_1 = 1$. In scenario 2, hour 1 must process
 1279 $2 - x$ units and A can take only one of them, so $\text{ALG}_2 = xp + 1 + (1 - x)\theta$,
 1280 while $\text{OPT}_2 = p + 1$. The ratio $r_1(x) = 1 + x(p - 1)$ increases in x , and the
 1281 ratio $r_2(x) = (1 + \theta - x(\theta - p))/(p + 1)$ decreases in x . Hence $\min_x \max\{r_1, r_2\}$
 1282 is attained where they are equal, at $x^* = (\theta - p)/(p^2 - p - 1 + \theta) \in [0, 1]$, and
 1283 its value is

$$1 + \frac{(p - 1)(\theta - p)}{p^2 - p - 1 + \theta}.$$

1284 The adversary chooses p to maximise this value, which gives $R^*(\theta)$. The
 1285 choice $p = \sqrt{\theta}$ gives the stated closed-form lower bound. For a randomised
 1286 algorithm, the expected cost in each scenario is affine in x , so it equals the
 1287 cost of the deterministic fractional algorithm that plays $\mathbb{E}[x]$, and the bound
 1288 applies against an oblivious adversary. Myopic MPC sees only the flexible job
 1289 at $t = 0$ and, for any $p > 1$, defers it entirely to the cheaper hour 1 ($x = 0$).
 1290 Its ratio in scenario 2 is $(1 + \theta)/(p + 1)$, which tends to $(1 + \theta)/2$ as $p \rightarrow 1^+$,
 1291 so its competitive ratio is at least $(1 + \theta)/2$. $\square$

1292 *Appendix A.2. Proof of Proposition 3 and Theorem 2*

1293 *Proposition 3.* A policy that completes all work W on time executes every
 1294 server-hour at a unit cost of at most U , so its emissions are at most UW .
 1295 Every server-hour in the oracle's solution costs at least L , and slack costs
 1296 $\Pi > U \geq L$, so $\text{OPT} \geq LW$. $\square$

1297 *Deadline safety.* We first show by induction that every pending real job j
1298 is window-feasible at the start of every hour: its remaining work satisfies
1299 $r_j \leq m_j(d_j - t)$. This holds at release by (A1). Suppose it holds at hour t ,
1300 and suppose the plan of CARMA leaves some real work $u_j > 0$ unfinished.
1301 Because $r_j \leq m_j(d_j - t)$, some hour $\tau \in [t, d_j)$ carries less than m_j units
1302 of job j in the plan. Moving a small amount of u_j to the backstop in hour
1303 τ keeps the plan feasible, because the backstop has unlimited capacity. It
1304 changes the planned cost by a negative amount, because the planner prices
1305 the backstop at most at $U < \Pi$ (A1), whatever its forecasts of the other costs
1306 are. Since $d_j - t \leq H$ (A1), hour τ lies inside the planning window. This
1307 contradicts optimality, so the plan completes all real work. After the first hour
1308 is executed, the rest of the plan is a feasible completion, so $r'_j \leq m_j(d_j - t - 1)$
1309 and the induction step holds. No real job is ever late, whatever the ghost
1310 jobs are, and θ -competitiveness follows from Proposition 3.

1311 *Telescoping.* Let σ_t be the state at the start of hour t : the remaining work of
1312 all real jobs released by t . Let $V_t(\sigma, G; c)$ be the optimal value of (2)–(5) over
1313 hours $t, \dots, T-1$ for the pending jobs σ and the future jobs G under costs c .
1314 Write $\bar{V}_t = V_t(\sigma_t, F_t; c)$, so that $\bar{V}_0 = \text{OPT}$ and $\bar{V}_T = 0$. For a feasible hour- t
1315 allocation x with cost $g_t(x)$ and successor state $\sigma^+(x)$, define

$$Q_t(x) = g_t(x) + V_{t+1}(\sigma^+(x), F_t; c), \quad \hat{Q}_t(x) = g_t(x) + V_{t+1}(\sigma^+(x), \hat{F}_t; \hat{c}_t).$$

1316 By dynamic programming over a deterministic future, $\min_x Q_t(x) = \bar{V}_t$, and
1317 $Q_t(x_t) = g_t(x_t) + \bar{V}_{t+1}$ for the executed action x_t . Summing over t gives

$$\text{ALG} - \text{OPT} = \sum_{t=0}^{T-1} \delta_t, \quad \delta_t = Q_t(x_t) - \min_x Q_t(x) \geq 0.$$

1318 Ghost jobs are released at $t+1$ or later, the planning window reaches T
1319 by (A2), and hour- t costs are observed. The planning problem solved by
1320 CARMA at hour t is therefore exactly $\min_x \hat{Q}_t(x)$, and x_t is a minimiser.

1321 *One-step loss.* Let $x^* \in \arg \min Q_t$. Since $\hat{Q}_t(x_t) \leq \hat{Q}_t(x^*)$, we have $\delta_t \leq$
1322 $\Delta(x_t) - \Delta(x^*)$, where $\Delta = Q_t - \hat{Q}_t$. Write $\Delta = D_1 + D_2$ with

$$\begin{aligned} D_1(x) &= V_{t+1}(\sigma^+(x), F_t; c) - V_{t+1}(\sigma^+(x), \hat{F}_t; c), \\ D_2(x) &= V_{t+1}(\sigma^+(x), \hat{F}_t; c) - V_{t+1}(\sigma^+(x), \hat{F}_t; \hat{c}_t). \end{aligned}$$

1323 *Cost error.* An optimal solution under $\hat{c}_t$ is feasible under c , and its cost
 1324 changes by at most ε_t per unit of flow on future-hour arcs. The same holds
 1325 with the roles of c and $\hat{c}_t$ exchanged. Hence $|D_2(x)| \leq \varepsilon_t W_t$ for all x , and
 1326 $D_2(x_t) - D_2(x^*) \leq 2\varepsilon_t W_t$.

1327 *Demand error.* Under (A1), the penalty attached to a window-feasible
 1328 job does not affect any optimal value, provided it is at least U . If a solution
 1329 leaves slack on such a job, the argument used for robustness moves that slack
 1330 to the backstop at no greater cost. True future jobs (penalty Π) and ghost
 1331 jobs (penalty Π_g) can therefore be treated as identical atoms whenever they
 1332 coincide. Let a be a window-feasible atom and G any set of future jobs. For
 1333 every state σ ,

$$Lw_a \leq V(\sigma, G \cup \{a\}) - V(\sigma, G) \leq Uw_a.$$

1334 For the upper bound, add a to an optimal solution for G by running it on the
 1335 backstop at rate $w_a/(d_a - a_a) \leq m_a$ in every hour of its window. For the lower
 1336 bound, delete a 's flow and slack from an optimal solution for $G \cup \{a\}$; each
 1337 deleted unit cost at least $\min\{L, \Pi_a\} = L$, since $\Pi_a \geq U \geq L$. Now transform
 1338 $\hat{F}_t$ into F_t by removing $\hat{F}_t \setminus F_t$ and adding $F_t \setminus \hat{F}_t$, one atom at a time. Then
 1339 $D_1(x)$ is a sum of increments, each lying in $[Lw_a, Uw_a]$ (with a minus sign
 1340 for removals) for every x . Therefore $D_1(x_t) - D_1(x^*) \leq (U - L) \sum_a w_a =$
 1341 $(U - L) d_t$.

1342 Combining the two bounds gives $\delta_t \leq (U - L)d_t + 2\varepsilon_t W_t$. Summing over t
 1343 and taking the minimum with the robustness bound proves (9). If $d_t = \varepsilon_t = 0$
 1344 for all t , then $\delta_t = 0$ and $\text{ALG} = \text{OPT}$. $\square$

1345 *Appendix A.3. Proof of Theorem 3*

1346 Write the problem of coalition C as $c(C) = \min\{c^\top z : A_+=z =$
 1347 $b_+(C), A_+z \leq b_+(C), z \geq 0\}$, where $z = (x, u)$. Jobs submitted out-
 1348 side C have zero work and zero width, and sites outside C have zero capacity.
 1349 This forces the corresponding flows to zero, so the problem coincides with
 1350 the stand-alone problem of C . The right-hand side is additive over sites:
 1351 $b(C) = \sum_{o \in C} b^o$, where b^o collects the work and widths of the jobs submitted
 1352 at o and the capacity of site o . The matrix A and the cost c do not depend
 1353 on C . The dual problem $\max\{b(C)^\top y : A^\top y \leq c, y_+ \leq 0\}$ therefore has a
 1354 feasible set that is the same for all coalitions.

1355 Let $y^* = (\lambda, \nu, \mu)$ be an optimal dual solution for $\mathcal{S}$. The slack variables
 1356 make every primal problem feasible, and costs are non-negative, so strong

1357 duality gives $c(\mathcal{S}) = b(\mathcal{S})^\top y^* = \sum_o (b^o)^\top y^* = \sum_o \phi_o$, which is budget balance.
 1358 For any coalition C , y^* is dual-feasible for C , so weak duality gives $c(C) \geq$
 1359 $b(C)^\top y^* = \sum_{o \in C} \phi_o$, which is the core condition [31]. Expanding $(b^o)^\top y^*$
 1360 gives (10). By Proposition 1 the problem is a min-cost flow, so optimal node
 1361 potentials, together with the reduced costs they induce on saturated capacity
 1362 arcs, also give an optimal dual solution. Any optimal dual solution can be
 1363 used; in our experiments it is returned by the LP solver HiGHS. $\square$

1364 **Appendix B. Forecast evaluation**

1365 Table B.1 and Fig. B.1a summarise forecast accuracy in 2024; hours
 1366 affected by gap filling are excluded. Averaged over horizons and sites, the
 1367 direct GBM forecaster has a mean absolute error (MAE) of 31.7 gCO₂/kWh,
 1368 against 35.3 for persistence and 40.9 for the seasonal-naive forecast. The
 1369 advantage is largest at short and medium horizons: MAE falls by 23% at
 1370 1 hour (6.4 against 8.3) and by 24% at 6 hours (23.6 against 31.2). The
 1371 advantage shrinks with the horizon, and at 24 hours the GBM is worse than
 1372 both benchmarks (42.7 against 40.9). Without weather forecasts, day-ahead
 1373 changes in wind output are hard to predict from the intensity history alone [44].
 1374 By site, the improvement over persistence is largest in absolute terms for
 1375 the volatile southern sites (West Midlands 50.2 against 57.7, London 38.0
 1376 against 43.4), although their MAE is also the highest. In South Scotland,
 1377 where intensity is low and often flat, persistence is better overall (14.5 against
 1378 16.2). Diebold–Mariano tests [68] with absolute-error loss and HAC variance
 1379 confirm these patterns. For the three English sites, the GBM is significantly
 1380 more accurate than persistence at horizons of 1–18 hours ($p \leq 0.002$) and
 1381 indistinguishable at 24 hours ($p \geq 0.09$). For South Scotland it is significantly
 1382 better at 1 and 3 hours, indistinguishable at 6 hours and significantly worse
 1383 from 12 hours onwards.

1384 Calibration matters more than point accuracy for the risk-adjusted cost
 1385 view, and Fig. B.1b shows the effect of distribution shift. Static split-conformal
 1386 bounds, calibrated on July–September 2023, under-cover in 2024 at every
 1387 level: at $\beta = 0.9$, annual coverage was 0.868 and monthly coverage ranged
 1388 from 0.78 to 0.95. With the adaptive recalibration, annual coverage is within
 1389 0.003 of the target at all levels (0.499, 0.598, 0.697, 0.797 and 0.898). Monthly
 1390 coverage stays within about four percentage points of the target (at most 4.2
 1391 points), and coverage is uniform across horizons. At $\beta = 0.5$, the recalibrated
 1392 median lies on average 10.3 gCO₂/kWh below the point forecast, which

Table B.1: Forecast accuracy on the 2024 test year (MAE in gCO₂/kWh; mean over sites unless stated).

Model	Horizon h (hours)					All horizons	
	1	3	6	12	24	MAE	RMSE
Persistence	8.3	20.0	31.2	38.6	40.9	35.3	49.2
Seasonal naive (24 h)	40.9	40.9	40.8	40.8	40.9	40.9	57.4
Direct GBM (this study)	6.4	15.1	23.6	34.5	42.7	31.7	41.6

MAE by site over all horizons (persistence / GBM): South Scotland 14.5 / 16.2; North West England 25.5 / 22.3; West Midlands 57.7 / 50.2; London 43.4 / 38.0.

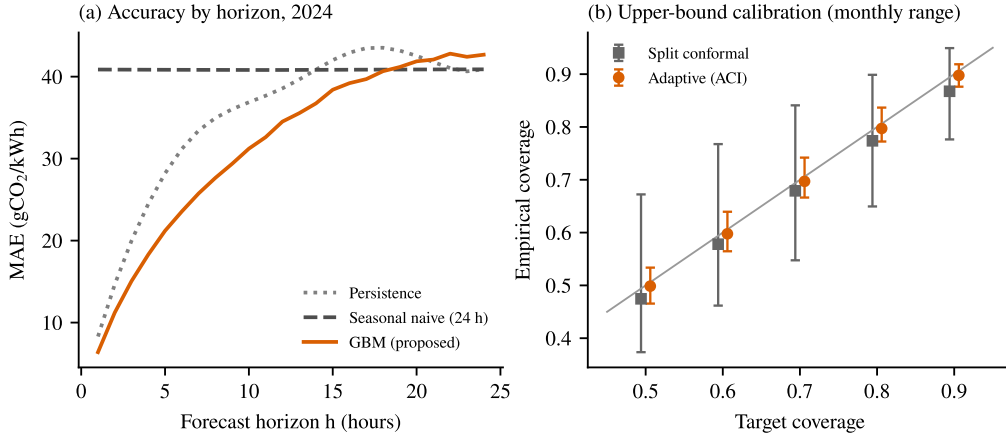


Figure B.1: Forecast evaluation in 2024. (a) MAE by horizon, averaged over sites. (b) Empirical coverage of one-sided upper bounds U^β against the target level β for static split conformal calibration and adaptive conformal inference (ACI). Markers show the annual coverage and bars the range across the 12 months.

reflects the downward drift in intensity that the static model does not capture. Coverage is not the whole story. In pinball loss, the adaptive bounds improve on static calibration at levels 0.5–0.7 (15.01 against 15.16 at 0.5) but are worse at 0.8 and 0.9 (9.07 against 8.36 at 0.9). Delayed feedback explains much of this. The level for horizon h is updated only when the outcome is observed, h hours after issue, so a day of under-forecasts drives the level of long horizons towards 1 before any correction arrives. At $\beta = 0.9$, 5.6% of the 2024 bounds (0.5% at $\beta = 0.8$) were at the fallback ceiling, the training-period maximum of the site. These features, together with the finite ceiling and the quantile grid, are why we treat the procedure as an empirical recalibration and do not invoke the coverage guarantee of ACI [54]. Annual coverage says nothing about conditional coverage in the site-hours that the optimiser selects.

Appendix C. Numerical check of Theorem 2

The weekly experiments of Section 5 violate the assumptions of Theorem 2, so we check the theorem separately in episodes that satisfy them (script `verify_theorem2.py`). Each episode is one day ($T = 24 = H$ hours) of the synthetic workload, with days starting at 26 test weeks, a backstop of unlimited capacity at the episode’s largest unit cost U (A1), window-feasible jobs, real-job penalty $\Pi = 10^4 > U$ and ghost penalty $\Pi_g = U$. Ghost jobs are the true future jobs, each omitted from the predictions with probability q (demand error), and the carbon forecasts of future hours receive i.i.d. Gaussian noise of standard deviation σ gCO₂/kWh (carbon error). For each episode, we compute the clairvoyant optimum OPT with the backstop, the emissions of CARMA, and the right-hand side of the smoothness bound (9) from the realised d_t , ε_t and W_t . There are 26 episodes for each of 12 settings (two loads and six error settings).

Three observations follow. With exact predictions, CARMA reproduces the optimum to within the 0.01 server-hour quantisation of the flow solver (relative difference at most 2×10^{-5}), which is consistency. No real work was late in any of the 312 episodes, in line with deadline safety. The smoothness bound holds in every episode but is loose by one to two orders of magnitude when predictions are wrong, because each misprediction enters d_t in every hour in which it is visible. The realised excess over the optimum is at most 37%, so the bound is a qualitative statement that errors enter linearly, not a tight prediction.

Table C.1: Numerical check of Theorem 2 in one-day episodes with a backstop. ALG/OPT: emissions of CARMA relative to the clairvoyant optimum (mean and maximum over 26 episodes). Bound: mean ratio of the right-hand side of (9) to OPT. The bound held in every episode, and no real work was late.

Load ρ	q	σ	Mean ALG/OPT	Max ALG/OPT	Bound / OPT
0.5	0	0	1.000	1.000	1.0
0.5	0	25	1.011	1.031	37.7
0.5	0	100	1.061	1.257	139.6
0.5	0.1	0	1.009	1.072	6.4
0.5	0.3	0	1.066	1.365	17.9
0.5	0.3	100	1.094	1.332	160.2
0.9	0	0	1.000	1.000	1.0
0.9	0	25	1.001	1.003	19.3
0.9	0	100	1.004	1.015	69.3
0.9	0.1	0	1.001	1.011	3.6
0.9	0.3	0	1.010	1.112	9.1
0.9	0.3	100	1.010	1.029	78.2

1428 **Appendix D. Results on other grids at all loads**

Table D.1: Results on five grids at all loads (migration allowed; 52 test weeks of 2024). Gaps to the oracle (%). Advantage: gap of the best baseline without demand information minus CARMA’s gap, with a 95% block-bootstrap CI (points). Late: CARMA’s share of late work (%).

Fleet	ρ	Oracle red.	MPC	MPC-Prot.	MPC+conf.	CARMA	Best baseline	Advantage (CI)	Late
Great Britain	0.3	71.8	19.1	28.8	15.5	9.3	MPC+conformal	6.3 (4.5–8.2)	0.00
	0.5	58.2	23.0	16.3	16.6	9.8	MPC-Protect	6.5 (5.2–7.3)	0.00
	0.7	41.1	20.2	17.2	16.0	8.0	MPC+conformal	8.0 (6.8–9.0)	0.02
EU	0.3	77.6	14.7	17.2	12.6	5.3	MPC+conformal	7.3 (6.1–8.8)	0.00
	0.5	63.0	19.7	13.0	17.5	5.8	MPC-Protect	7.3 (6.2–8.7)	0.00
	0.7	43.6	13.9	12.5	13.1	5.5	MPC-Protect	7.0 (6.2–8.1)	0.03
US	0.3	54.3	11.1	17.5	9.8	5.6	MPC+conformal	4.2 (3.5–5.1)	0.00
	0.5	34.6	7.0	7.7	6.7	4.3	MPC+conformal	2.4 (2.0–2.9)	0.00
	0.7	21.8	5.2	5.5	5.0	3.4	MPC+conformal	1.7 (1.4–2.0)	0.00
BR	0.3	46.5	4.3	8.3	4.2	2.9	MPC+conformal	1.3 (0.8–1.7)	0.00
	0.5	37.5	6.6	6.0	4.4	2.7	MPC+conformal	1.7 (1.2–2.3)	0.00
	0.7	28.0	6.7	6.7	5.0	3.3	MPC+conformal	1.7 (1.2–2.4)	0.01
GLOBAL	0.3	66.6	6.9	9.0	5.5	3.6	MPC+conformal	1.9 (1.3–2.6)	0.00
	0.5	56.0	9.2	6.8	7.2	4.0	MPC-Protect	2.8 (2.4–3.3)	0.00
	0.7	42.6	13.1	10.7	11.7	6.0	MPC-Protect	4.7 (3.8–5.6)	0.00

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